



Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflameChemical kinetics modeling of *n*-nonane oxidation in oxygen/argon using excited-state species time historiesBrandon Rotavera^{a,*}, Philippe Dagaut^b, Eric L. Petersen^a^a Department of Mechanical Engineering, Texas A&M University, College Station, TX 77843, USA^b Institut de Combustion Aérodynamique Réactivité et Environnement (ICARE), Centre National de la Recherche Scientifique (CNRS), 45071 Orléans Cedex 2, France

ARTICLE INFO

Article history:

Received 20 July 2013

Received in revised form 9 November 2013

Accepted 10 November 2013

Available online xxx

Keywords:

OH^{*} time histories

Chemical kinetics modeling

Shock tube

n-Nonane

Ignition delay time

ABSTRACT

Chemical reactions of ground-state species strongly govern the formation of excited-state species, including OH^{*} and CH^{*}, which are commonly used to determine ignition delay times of fuels. With well-characterized chemiluminescence rates embedded in chemical kinetics mechanisms, time histories of excited-state species can aid in identifying influential ground-state reactions which are important to processes such as ignition delay time. Placing emphasis on the high-temperature regime, improvements were made to a detailed chemical kinetics mechanism of *n*-nonane oxidation developed previously by the authors. Using characteristic features of OH^{*} time histories measured in shock-tube experiments as a metric, detailed model analyses were performed over a broad range of conditions: $T > 1100$ K, $1.5 < P$ (atm) < 10.5 , $\phi = 0.5, 1.0, 2.0$. OH^{*} time history measurements, particularly under fuel-rich conditions ($\phi = 2.0$), displayed a two-peak behavior, with the first peak occurring within the first 5–10 μ s after reflected-shock passage, and the second, wider peak corresponding to main oxidation and ignition. In the initial version of the kinetics mechanism, the two peaks at rich conditions were predicted to merge, blurring the main ignition process prior to the second peak. The work herein presents modifications to the initial chemical kinetics mechanism which led to improved agreement between measurements and model-based predictions, with emphasis on the fuel-rich condition. To this end, the predicted shapes of the OH^{*} time histories were crucial to matching the two-peak behavior detected in the experiments. A first-order resistance-capacitance (RC) model of the experimental time response of the optical setup was developed and shown to reproduce the measured time dependence and peak behavior that are vital for matching the OH^{*} behavior near time-zero. The RC model processes the kinetics predictions in a way that allows the kinetics model predictions to directly correspond to the true conditions in the experiment. In moving towards improved agreement in OH^{*}-profile predictions for all conditions, improvements in the kinetics mechanism were also realized at the two leaner equivalence ratios ($\phi = 1.0$ and 0.5), both in terms of OH^{*} profile shape and ignition delay times. Model calculations of oxidation processes indicate that reactions leading to the first OH^{*} peak originate from fuel homolysis. The resulting (alkyl) radicals lead to the formation of methyl which then, through a series of H-abstraction reactions, leads to the production of the methylidyne radical (CH) that reacts with molecular oxygen to form OH^{*}. The oxidation processes near time-zero terminate, in part, due to methyl depletion by methylene forming $C_2H_4 + H_2$. In addition to the insight gained on *n*-nonane ignition and oxidation chemistry, the present study highlights the utility of correctly interpreted OH^{*} measurements for inference of kinetic information other than ignition delay times.

© 2013 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

Fundamental combustion chemistry measurements of higher-order hydrocarbons are necessary when developing chemical kinetics mechanisms for transportation fuels. Such measurements should ideally span a wide parameter space (temperature, pressure, equivalence ratio, and fuel concentration). With *n*-alkanes

commonly occupying a large volumetric percentage of spark-ignition, compression-ignition, and aviation fuels, accurate modeling of engine-relevant combustion processes for *n*-alkanes is needed [1]. Simulations by Westbrook et al. show that higher-order *n*-alkane ignition delay times are highly similar at temperatures in excess of 1000 K [2] – a result supported by experimental evidence [1,3–5]. Several chemical kinetics mechanisms for higher-order *n*-alkanes have been developed [2,6–13], yet improvements are still required for wide-ranging accuracy in ignition delay time predictions. Chemical kinetics measurements including species time

* Corresponding author.

E-mail addresses: brotave@sandia.gov, rotavera@aggienetwork.com (B. Rotavera).

histories, in addition to ignition delay times, are useful for improving the accuracy and applicability of chemical kinetics models.

Laser-based absorption measurements of ground-state species [14–17] and other forms of speciation [18,19] are common and afford stringent, quantitative experimental targets. In addition, chemiluminescence also serves as a versatile diagnostic for numerous combustion-related phenomena. In gas turbine engines, OH^* and CH^* chemiluminescence has been used as a metric for equivalence ratio measurements, where control over equivalence ratio fluctuations was achieved using signal intensity feedback from CH chemiluminescence in methane–air combustion under engine-relevant conditions [20,21]. Imaging of chemiluminescence has been utilized to perform laminar flame speed and Markstein-length measurements of spark-ignited propane–air flames at 1 atm [22] and has also been employed to determine autoignition times inside practical engine configurations [23]. In addition to measurements for practical applications, chemiluminescence also has utility in fundamental modeling of chemical kinetics [1,3,24–26], where shock-tube measurements of excited-state hydroxyl (OH^*) radicals, methylidyne (CH^*) radicals, or other excited species aid in the evaluation of the accuracy of chemical kinetics mechanisms. In shock-tube experiments, the chemiluminescence signal is recorded as a time history, or profile, which is directly proportional to the concentration of the excited species that is producing the light emission. Such profiles serve as an additional diagnostic for validation of chemical kinetics mechanisms by providing two important chemical kinetics-related targets: ignition delay times and peak yield times. However, chemiluminescence time histories are, arguably, under-utilized for providing kinetic information other than characteristic times.

The impetus for the present work centered around the use of OH^* and CH^* chemiluminescence to improve a chemical kinetics model by re-evaluating rate parameters of certain ground-state reactions identified as being important for the shape of excited-state species time histories. Specifically, chemiluminescence time history measurements of excited-state OH and complementary model calculations were employed to identify key reactions influential on both ignition delay times and the rates of OH^* formation and depletion for the purpose of refining a chemical kinetics model for n -nonane oxidation developed previously by the authors [6].

Before discussing the approach and results of the present study, the unique behavior in the OH^* time histories from the previous n -nonane study [6] and the inability of the initial mechanism to adequately reproduce the profiles, are reviewed. In the initial version of the model, discrepancies were present under certain conditions in time-dependent, excited-state OH^* predictions. In the shock-tube experiments reported in Rotavera et al. [6], OH^* formation was evident immediately after the reflected-shock passage (referred to herein as time-zero), particularly for fuel-rich conditions at elevated temperatures. Hence, two distinct peaks were observed in the OH^* profiles: the first peak near time-zero and the second peak appearing post-ignition (Fig. 1). Experimentally, the first OH^* peak was observed to be short-lived ($<75\ \mu\text{s}$) and smaller in magnitude than the main ignition peak. The chemistry related to the formation of the first OH^* peak and in relation to the ignition peak are the subject of the present study. As such, the present work demonstrates an often underutilized approach toward tuning chemical kinetics mechanisms – namely, the use of unique features in chemiluminescence time histories to tune the chemical kinetics of the ground-state species. Ignition delay time is defined herein as the intersection of the steepest rise on the second (ignition) peak with the initial concentration of OH^* , as shown in Fig. 1.

While ignition delay times and OH^* time histories were reproduced well for lean ($\phi = 0.5$) and stoichiometric equivalence ratios, model predictions [6] of the OH^* behavior at $\phi = 2.0$ were inadequate for the following two reasons. First, a dramatic over-predic-

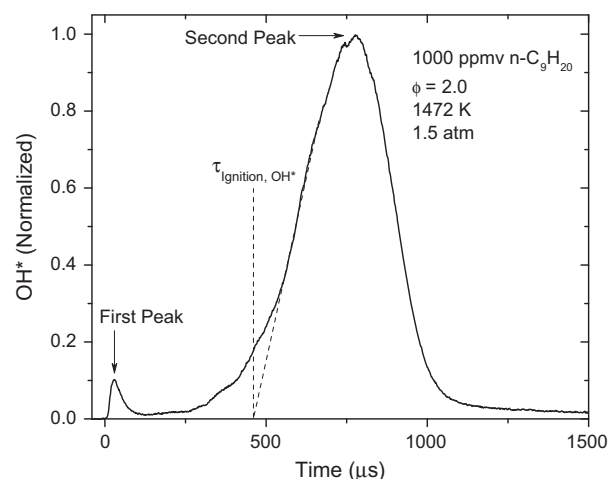


Fig. 1. Representative OH^* time history from the dataset utilized in Rotavera et al. [6] showing the OH^* peak near time-zero, occurring over a (measured) period of approximately $65\ \mu\text{s}$, relative to the main ignition peak. The ratio of the two OH^* peaks (first/second) is utilized herein as a tuning parameter for model improvement through analysis of ground-state chemical reactions leading to OH^* . Ignition delay time is defined using the method of steepest ascent.

tion of the first OH^* peak by more than two orders of magnitude resulted in a merging of the two peaks causing ignition delay times at the fuel-rich condition to be undefinable due to unclear ignition delay time (Fig. 2a). Second, model-predicted full-width half-maximum (FWHM) times of OH^* profiles were significantly wider compared to results measured for the $\phi = 2.0$ condition (cf. Fig. 2a).

The measured ratio of the first-to-second OH^* peaks, defined herein as η , is quantifiable and displays a dependence proportional to increasing temperature. However, the initial model calculations indicated an inverse relationship between η and temperature (Fig. 2b), with η becoming larger with lower temperature. For the lowest temperature covered experimentally at $\phi = 2.0$, 1.5 atm (1412 K), initial model calculations predicted the first peak to be 200 times larger than that of the main OH^* ignition peak (Fig. 2b). The dependence of OH^* peak ratio on temperature serves as an important target herein for model improvement.

Resulting from the discrepancy between model predictions and experiment outlined above, three primary issues were identified as targets for improving the ability of the initial mechanism to model excited-state OH time histories: (1) over-predicted peak ratios, i.e. $\eta \gg 1$; (2) broader model-predicted, time-dependent OH^* profiles post-ignition (i.e. broader FWHM relative to measurements); (3) undefinable model-predicted ignition delay times for fuel-rich conditions. Discussed in the ensuing sections are details of the chemical kinetics model, including modifications, experimental conclusions on the time-zero OH^* formation observed during n -nonane oxidation, sensitivity analysis, reaction pathway analysis of n -nonane oxidation near time-zero, and details on the methodology leading to the improved predictive capability of the mechanism for ignition delay times. Ignition delay time measurements from Rotavera et al. [6] and Rotavera and Petersen [1], along with measurements taken at 4% O_2 for the present study, are compared to model predictions using both the initial mechanism [6] and the modified mechanism herein. Rate-of-production analysis conducted on CH^* is also discussed to explain the absence of the species at times immediately after shock passage.

2. Chemical kinetics model

The mechanism herein utilizes strong hierarchical structure designed to model both low- and high-temperature chemistry.

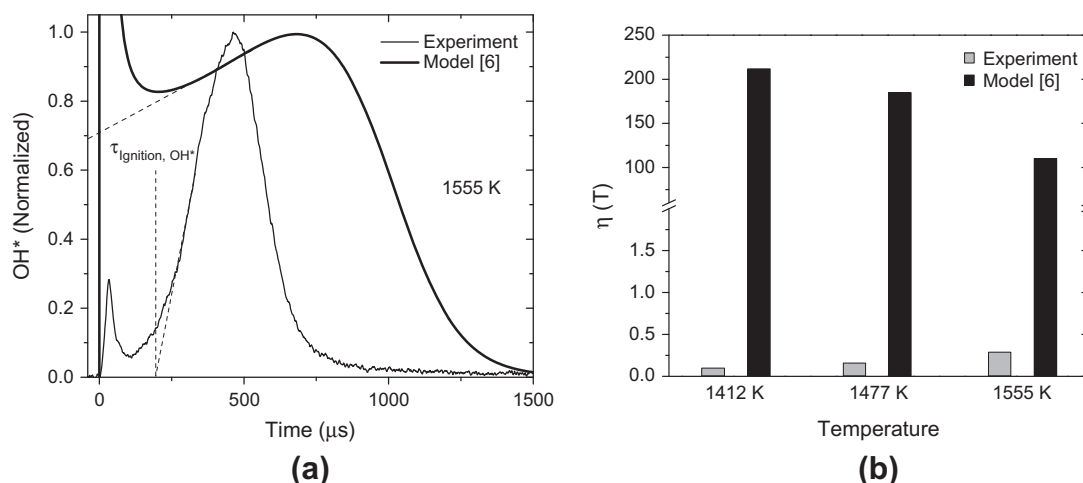


Fig. 2. (a) Initial model prediction of a representative OH^* time history normalized to the main ignition peak during fuel-rich oxidation of n -nonane Ar (1000 ppmv, 1555 K, 1.5 atm). Despite definable ignition delay times at other conditions, the initial version of the mechanism [6] did not permit definition of ignition under $\phi = 2.0$ conditions due to over-prediction of OH^* near time-zero. Using the standard definition based on steepest slope (indicated by dashed lines) model extrapolation led to negative time. (b) Temperature trends of experimental and modeled [6] first-to-second peak OH^* ratios, denoted $\eta(T)$, showed strong disagreement.

Rate coefficients come mainly from Buda et al. [27]. The high-temperature reactions primarily consist of C–C and C–H bond scissions from n -nonane and nonyl radicals. The OH^*/CH^* subset of Hall and Petersen [28] was integrated into the mechanism to model chemiluminescence profiles and determine ignition delay times. The present mechanism consists of 729 species and 3445 reactions (refer to [Supplementary material](#)). Thermodynamic coefficients were obtained from the Burcat and Ruscic database [29] when available, or were estimated using the THERGAS software [30] based on the well-known group additivity methods proposed by Benson [31]. Rate coefficients for reverse reactions were computed from the corresponding forward rate coefficients and the appropriate equilibrium constant calculated from thermochemical data.

Kinetic modeling of OH^* and CH^* profiles and detailed analyses were performed using ChemKin PRO v15101, where species profiles from the shock-tube experiments were calculated using the homogeneous batch reactor module. Under conditions of sufficiently high dilution (>97% Ar), the constant-enthalpy/constant-pressure constraint was implemented to simulate the thermodynamic environment within which the reactions occurred. Due to the slight rise in pressure, post-reaction, resulting from the liberation of heat at the 4% oxygen condition, the constant-internal energy/constant-volume module constraint of the homogeneous batch reactor was used to model those chemiluminescence profiles, taking into account the shock-tube specific dP/dt rate (discussed in detail in subsequent sections).

3. Experimental methodology

In the present study, chemiluminescence profiles obtained during experiments from [6] were the primary focus, with respect principally to model-based determination of ground-state reactions controlling OH^* time dependence. In addition to profile analysis of excited-state species, the present work reports new experiments on n -nonane ignition delay times at fixed oxygen concentration of 4% (vol.); measurements from [6] involved fixed fuel concentration (1000-ppmv), with oxygen ranging from 0.7% to 2.8%. Experimental details are summarized briefly here since these aspects form the basis of the present modeling and experimental interpretations. A subsequent section outlines the optical considerations in the present study.

3.1. Shock tube facility

Two shock-tube facilities were employed to measure species time histories and ignition delay times. Experiments were conducted primarily at highly diluted conditions (>97% diluent), with some additional ignition measurements taken at a fixed oxygen concentration of 4% (vol.), corresponding to ~95% Ar dilution, to compare with other n -nonane ignition delay time results in the literature, namely Davidson et al. [3]. Table 1 provides volumetric percentages, corresponding equivalence ratios, and reflected-shock pressures used in the measurements reported herein. The experimental methodology and details of the facilities are described in Rotavera [32], de Vries et al. [33], and Aul et al. [34]. In brief, for 1.5-atm experiments, measurements were made using a 4.0-m long, stainless steel shock tube (117 cm²) described in detail by Rotavera [32], and elevated-pressure experiments (10 atm) were conducted using a 4.7-m long stainless steel shock tube with a 16.2-cm inner-diameter (256 cm²) [33]. Mixtures were prepared manometrically wherein the liquid-phase n -C₉H₂₀ was exposed to a high-vacuum (<1 μtorr) mixing chamber and permitted to evaporate to approximately 50% of its vapor pressure at 300 K (ca. 1.5 torr). A perforated tube (stinger) extending the length of the mixing chamber(s) allowed the gases to enter through numerous tiny, turbulent jets which induce rapid mixing. Results were highly reproducible using the same procedures in both facilities, where measured ignition behavior showed consistent agreement as evidenced by overlapping ignition delay time results.

Table 1

Percent-volume composition, corresponding equivalence ratios and average reflected-shock pressures. Pressure refers to the pressure immediately after passage of the shock wave (i.e. time-zero), and represents the averaged pressure from the group of experiments in the ranges shown (i.e. near 1.5 and 10 atm).

ϕ	n -C ₉ H ₂₀ (%)	O ₂ (%)	Ar (%)	P (atm)	Reference
0.5	0.10	2.8	97.1	1.5, 10.4	[6]
	0.14	4.0	95.9	1.5	Present Study
1.0	0.10	1.4	98.5	1.5, 10.4	[6]
	0.29	4.0	95.7	1.5	[1]
2.0	0.10	0.7	99.2	1.5, 10.3	[6]
	0.57	4.0	95.4	1.5	Present study

As shown in Table 1, the Ar dilution levels ranged from 95.4 to 99.2% of the mixture by volume. Shock-tube experiments performed using high levels of dilution (>97% diluent concentration by volume) yield favorable thermodynamic conditions from which well-controlled measurements of chemiluminescence time histories are extracted. The high dilution mitigates reaction exothermicity and renders the conditions behind the reflected shock wave effectively constant within the timeframe of the experiment (i.e. <2 ms). This timeframe of nearly constant conditions is restricted to shock tubes with sufficiently large inner diameters, where boundary layer growth does not appreciably occupy a large volumetric percentage of the reacting gases. Resulting from non-ideal gas dynamic effects, however, due to rate-of-pressure increases, conditions behind reflected shock waves can vary slightly with time, yet can be modeled accurately using the measured reflected-shock pressure and an imposed linear fit of the pressure change extended to the test time of the experiment.

Although slight in the present work, existing in all shock-tube facilities due to non-ideal gas dynamic effects are rate-of-pressure increases which depend highly on shock-tube-specific geometry, experimental test temperature and pressure, and diluent species. It is important to determine the time-dependent pressure behavior of shock tubes to accurately report the temperature at which reaction occurs, in addition to accounting for temperature variation with time when performing model calculations. Increased pressure variation with time (dP/dt) relative to initial, reflected-shock pressures were characterized in the abovementioned shock-tube facilities at nominal rates of 3.0%/ms for the 10.9-cm inner-diameter square shock tube [32] and 2.1%/ms for the 16-cm inner-diameter shock tube [33], leading to negligible rate-of-temperature increases (relative to the reflected-shock temperature) of 4.3%/ms and 1.3%/ms, respectively using the assumption of an isentropic process post reflected-shock. To assess the impact of the dP/dt behavior on modeling of the chemical kinetics, calculations were carried out using a linear rate-of-pressure increase for the most extreme case observed in the present work (3%/ms) for a 98.5% Ar dilution level. Comparison is drawn against the isobaric case relative to the 3%/ms condition (Fig. 3b). No significant impact either on the profile shape or the measured ignition delay time is observed.

Experiments were conducted in Rotavera et al. [6] behind reflected shock waves over the temperature range $1176 < T$ (K) < 1653 wherein excited-state hydroxyl ($X^2\Pi \leftarrow A^2\Sigma^+$ transition

of OH) and excited-state methylidyne ($X^2\Pi \leftarrow A^2\Delta$ transition of CH) radical time histories were measured using a Hamamatsu 1P21 photomultiplier tube (PMT). Such profiles provide several features of utility for modeling of the associated chemical kinetics including species time dependence, ignition delay times, and peak yield times. Sidewall-monitored measurements of OH* or CH* are used to define ignition delay time by the method of steepest ascent, wherein the point of intersection of a line drawn along the steepest gradient in the ignition marker (chemiluminescence) and the line of zero-concentration meet defines the time of ignition (Fig. 3a). Uncertainty in ignition delay times reported herein is due primarily to uncertainty in shock velocity measurements and the subsequent reflected-shock temperature calculations. The error in test temperature has been characterized for both facilities to be ± 15 K using the methodology of Petersen et al. [35]. In addition to characteristic times, which provide macroscopic targets, chemiluminescence profiles also serve as time-dependent chemical kinetics targets given the inherent dependence on ground-state chemistry of excited-state radical production. Note in Fig. 3a the appearance of OH* immediately upon formation of the reflected-shock conditions (i.e. time-zero). The chemistry forming the first peak in OH* is oxidative in nature and due initially to the formation of alkyl radicals from rapid fuel decomposition and subsequent oxidation chemistry. The magnitude of the first peak relative to the second OH* peak, occurring post-ignition, forms the basis of the kinetics analysis in the present work.

3.2. Optical setup considerations

Comparison of the time behavior of OH* profiles in Fig. 3b indicates that the experimental measurements are broader (in time) relative to model-based predictions. Differences increased towards lower temperatures based on inspection of all experimental data. Model-predicted time histories that possess a more-narrow FWHM relative to experimental profiles were reported in other studies as well [36,37]. In general, such differences can be due to deficiencies in the kinetics mechanism or to some experiment-driven, artificial broadening effect. In a system where the ground-state kinetics are well-known, such as for H_2-O_2 , the time histories of excited-state species such as OH* agree strongly with model predictions [36,37]. With regard to experiment-induced broadening, such phenomena can come from at least two different sources in shock-tube experiments: (1) passage of the reflected shock wave across the

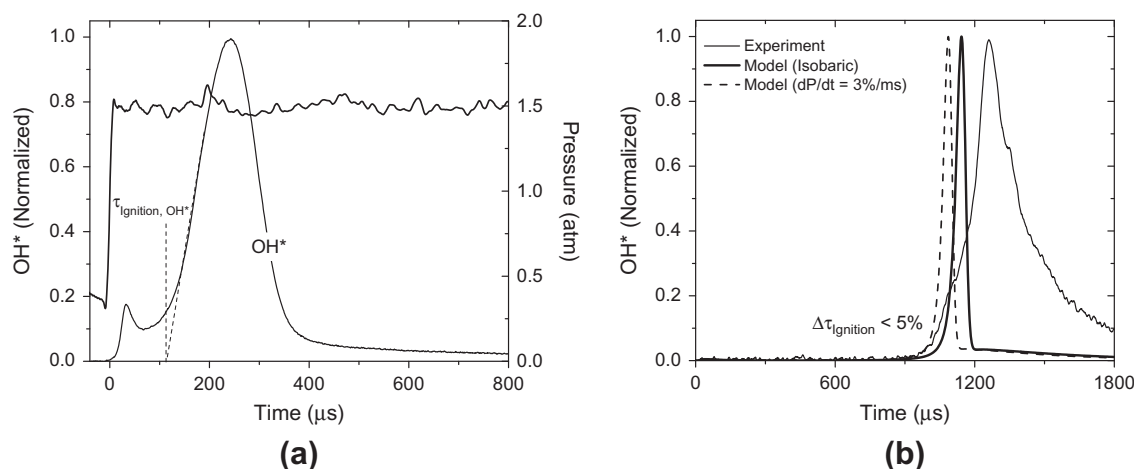


Fig. 3. (a) OH* chemiluminescence time history showing definition of ignition delay time and OH* formation near time-zero resulting from immediate fuel decomposition and subsequent oxidation processes; $\phi = 2.0$, 99.2% Ar, 1614 K, 1.5 atm, $\tau_{\text{ignition, OH}^*} = 124 \mu\text{s}$. (b) OH* chemiluminescence time history modeled using the present mechanism under isobaric conditions and for a linear-rate-of-pressure increase of 3%/ms. Ignition delay times differ negligibly, indicating near-ideal reactor conditions in the experiments; $\phi = 1.0$, 98.5% Ar, 1447 K, 1.6 atm, $\tau_{\text{ignition, OH}^*} = 378 \mu\text{s}$.

path of the optical setup, which mainly affects signals around the time of shock wave passage near time-zero, and (2) the alignment and placement of the optical setup can be such that the light that reaches the detector can include emission from regions in the tube other than the intended location at the window plane. In a recent study by Davis and Petersen [38], the second phenomenon was thoroughly explored for the conventional optical setup employed herein, namely a narrow slit, a focusing mirror, and a PMT detector. Davis and Petersen [38] concluded that the measured OH* time histories are not appreciably sensitive to any reasonable perturbations in optical configuration.

Taking into context the above results, the broadening phenomenon of most interest to the study herein is that due to the passage of the reflected shock wave near time-zero. The effect is of possible concern for the study herein given the focus on measured signals near time-zero, for the first OH* peak. The broadening is dependent on the time it takes the shock wave to pass through the region being probed by the optical setup, as in Eq. (1). For reflected-shock speeds S_R on the order of 0.5 to 1 mm/ μ s in the stationary laboratory frame, the resulting broadening time could be on the order of a few microseconds for the setup used herein, with a 400- μ m slit width resulting in a focal spot (i.e. source) width of a few millimeters in the reaction zone.

$$t_{\text{Broad.}} \propto \text{Source Width}/S_R \quad (1)$$

The response time of the detector is also critical for capturing the rapidly changing signals near time-zero. Inherently coupled to chemiluminescence measurement is a dependence on the characteristic time response related to electronic acquisition and processing. To control the output voltage from the PMT detector electronics, an externally applied resistance is integrated into the signal output channel. The imposition of resistance results in a characteristic time response that depends on the level of applied resistance, R , and circuit capacitance, C . The added resistance increases the output voltage signal, yet also increases the RC time constant of the output signal. With time-response being the dominant factor in chemical kinetics measurements, specific attention was given to the effect of resistance on time response to ensure that processes near time-zero were captured accurately. For the experimental results discussed herein, the PMT was equipped with a signal output resistance of 4.8 k Ω . To remain within the region of linearity of the PMT detector, output voltages were limited to levels less than 500 mV (from previous calibration).

To determine whether the first-peak OH* signals were captured completely, in terms of peak level of OH* near time-zero and therefore accurate time response, the resistance applied to the PMT was altered from the 4.8-k Ω level used in [6] to 50 Ω , 470 Ω , 1 k Ω , or 10 k Ω , where lower resistance improves time response (from the smaller time constant) and higher resistance increases signal output. The latter three measurements of PMT time-response dependence to applied resistance is discussed in detail in Rotavera and Petersen [1]. The 50- Ω results obtained herein are compared to measurements in [1] in Fig. 4. Employing the different resistors, $\phi = 2.0$ *n*-nonane ignition experiments were repeated for the present study at the same conditions employed in [1] and [6], and a strong effect on the time constant was observed. Qualitatively, from a series of measurements near 1600 K using a fixed optical setup, lower levels of external PMT resistance produced higher magnitudes of OH* immediately after shock passage (Fig. 4), yet timescales were relatively unaffected (Fig. 4, inset). Note however that there is only a slight difference in peak OH* between the 50- Ω and 470- Ω cases (other than a decreased S/N), indicating that the 470- Ω resistance is nearly sufficient to capture most, if not all, of the full OH* peak near time-zero while still providing measurements with a high signal-to-noise ratio.

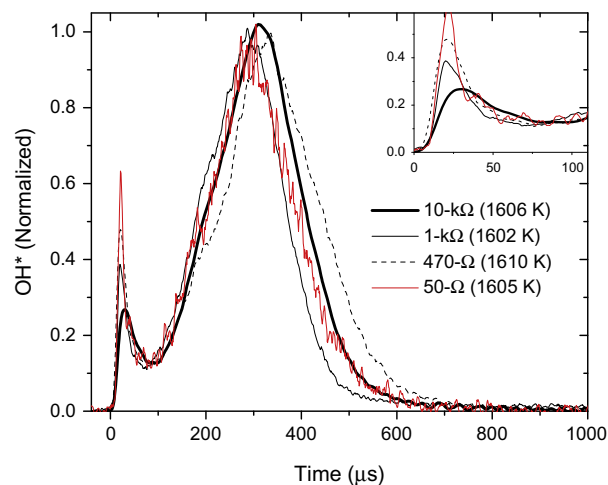


Fig. 4. OH* profiles during *n*-nonane oxidation near 1600 K, 1.5 atm in 99.2% Ar using four values of PMT output resistance: 50 Ω , 470 Ω , 1 k Ω , 10 k Ω . Lower applied PMT resistance causes larger incipient OH* peaks from increased time resolution. The slight delay in time response near time-zero is not important for the primary emphasis herein (and is addressed in more detail in subsequent sections). Refining the model-predicted first-to-second peak ratio is the purpose of the OH* profiles, not the specific concentration of OH* at incipient times.

Of significance to note is that the slower time response exhibited by higher-resistance measurements of chemiluminescence, in conjunction with any broadening due to the passage of the shock wave, impacts only the first ~ 30 μ s of the experiment, and the time dependence over the span of the measurement post-ignition is unaffected, maintaining highly reproducible ignition delay times. In determining the resistance dependence of the chemiluminescence profile measurements, it is important to note that the time dependence can be captured without electronic delay by using matched impedance with the measurement electronics (50 Ω). With respect to use of the OH* profiles as modeling targets, the primary emphasis remains unaffected: refining the model-predicted first-to-second OH* peak ratio, not the specific concentration of OH* at incipient times. Discussed in the following section is a simple model of the time response of the optical setup.

3.3. Model of optical diagnostic time response

From inspection of Fig. 4, it is observed that the effect of different detector electronics resistances is similar to the first-order response of an RC circuit from electronics or control theory. To illustrate the resistance effect and to derive a simple model for the response of the experiment in capturing the first peak, the following analysis was performed. The OH* concentration (i.e. measured voltage) was approximated by the triangular-type wave shown in Fig. 5. Upon inspection of the detailed kinetics predictions of OH* time histories, the initial OH* formation near time-zero takes about 10 μ s to reach the peak, and this time is represented in Fig. 5 as t_p . The normalized peak is shown on a 0-to-1 scale in the form of V/V_p , or the measured voltage V (i.e. concentration) divided by the peak voltage, V_p . From RC circuit theory, the time constant τ is defined as $\tau = RC$, and the system response to the ramp rise from $t = 0$ to $t = t_p$ is described by the solution [39]:

$$\frac{V^+(t)}{V_p} = \frac{t}{t_p} - \frac{\tau}{t_p} (1 - e^{-t/\tau}) \quad \text{for } t \leq t_p \quad (2)$$

$$\frac{V^+(t)}{V_p} = 1 - \frac{\tau}{t_p} (e^{t_p/\tau} - 1) e^{-t/\tau} \quad \text{for } t \geq t_p \quad (3)$$

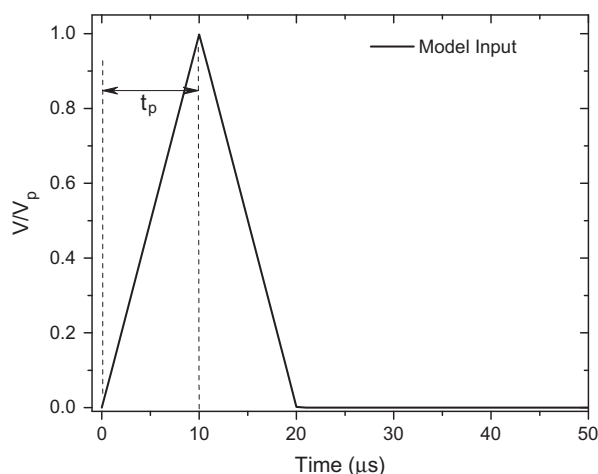


Fig. 5. Model voltage input to mimic the first OH* peak near time-zero utilized as input to the simple RC-circuit model of the optical system time response.

The superscript + refers to the system response to the ramp rise. By analogy, the ramp fall can be modeled as follows, with V^- referring to portion of the signal with the negative slope:

$$\frac{V^-(t)}{V_p} = \left(1 - \frac{t}{t_p}\right) + \frac{\tau}{t_p}(1 - e^{-t/\tau}) \quad \text{for } t_p \leq t \leq 2t_p \quad (4)$$

$$\frac{V^-(t)}{V_p} = \frac{\tau}{t_p}(e^{t_p/\tau} - 1)e^{-t/\tau} \quad \text{for } t \geq 2t_p \quad (5)$$

For times in the output signal greater than t_p , the solutions to the two responses are combined by taking the product of $V^+(t)$ and $V^-(t)$.

Using the measured results in Fig. 4 as a guide, the above equations were used to calculate the predicted response of the optical system to the Fig. 5 input signal for various capacitance values, C , to determine the best fit to the actual experimental setup. The result of the calculations was a value of $C = 4500$ pF which best fits the trends observed for the different resistance values from the excursion tests shown in Fig. 4. Figure 6 shows the calculated system response using the simple model with the known experiment resistances and the estimated value for C . Comparing Fig. 6 to the inset of Fig. 4 indicates that the trends are well-reproduced, certainly within the accuracy of the data and related S/N produced using the smaller resistance values. Note that the t_p is modeled well for the 10-k Ω resistance, for which the RC time constant is 45 μ s. Also shown in Fig. 6 is the additional curve for $R = 4.8$ k Ω ($\tau = 21.6$ μ s), the resistance employed for the majority of the experiments utilized herein. Note also that the 50- Ω resistance ($\tau = 0.2$ μ s) should capture the true first peak kinetics near perfectly, per the RC-model results in Fig. 6.

The end result of the simple RC model is used as a filter for calculations using the chemical kinetics mechanism. In other words, the mechanism predictions can be run through the simple RC model to produce a time dependence that can be compared directly to the experiments. Such a comparison is shown in subsequent sections, with the net result that validates the model improvements shown in present work, that were based on fitting the two peaks from the measured OH* time histories.

4. Chemical kinetic modeling of chemiluminescence profiles

Formation of excited-state species is controlled by both collisional energy transfer processes and ground-state chemical reactions, with the latter shown in previous work by Hall and

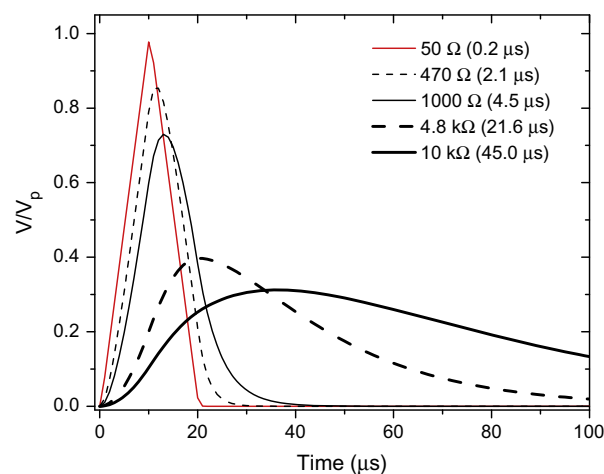


Fig. 6. Predictions of the optical system response to the first OH* peak input (Fig. 5) using the RC-circuit model presented in the text. The various resistances correspond to the resistances used in the measurements presented in Fig. 4, with the addition of the 4.8-k Ω case; RC time constants for respective resistances are shown in parentheses.

Petersen [28] to be more dominant in the case of OH* formation. Chemiluminescence from OH* observed immediately upon formation of the reflected shock waves appeared in the majority of the highly diluted *n*-nonane measurements in the previous experimental study [6]. The primary interest herein is the formation of OH* near time-zero prior to, and independent of, the primary ignition event. The behavior is ascribed to immediate fuel decomposition and subsequent oxidation of radicals which in turn, through several possible pathways, produce CH. Since the dominant OH* production channels involve ground-state chemistry such as CH and O₂ [28], measurements of time-dependent chemiluminescence profiles may be regarded as kinetically relevant for the purposes of serving as a tuning parameter for chemical kinetics models in the context of identifying ignition-controlling reactions.

In the present work, observations on the magnitude of the first OH* peak relative to peak OH* formation resulting from the main oxidation process are encompassed by the variable η , defined explicitly in Fig. 7. The phenomenon causing the first OH* peak (near time-zero) showed strong dependence on several experimental parameters; in descending order of impact: temperature, equivalence ratio, fuel concentration, and pressure. Experimentally, the influence of individual parameters on η were elucidated by conducting experiments using variation in the individual parameter while holding all other experimental conditions constant. Therefore, the ratio η serves as a characteristic feature of the chemiluminescence profiles which can then be used quantitatively to study the influence of ground-state reactions on its value.

The two-peak behavior in OH* is captured by the model from [6], yet at conditions identical to those in a given experiment, the first OH* formation peaks were highly overpredicted. In the experiments, using 4.8-k Ω PMT resistance, the time-zero peaks rose to, at most, 40% of the peak OH* formation post-ignition. Maximum time-zero peaks occurred under fuel-rich conditions at high temperatures. As a quantitative comparison between model prediction and experiment, model computations from Rotavera et al. [6] show that at 1591 K, 1.5 atm, predicted OH* at the first peak rises to a value 110 times greater than the OH* peak post-ignition, while the experimental result was 0.17 (i.e. $\eta \gg 1$ from model calculations and $\eta < 1$ experimentally). Although predictions of the magnitude of η were highly over-pronounced, the initial kinetics model [6] does agree with the several noted experimental trends relevant to the height of the OH* peaks near time-zero. For example, the

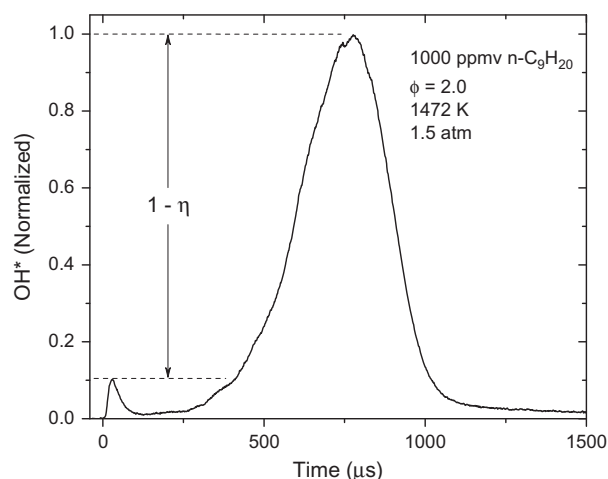


Fig. 7. Definition of first-to-second OH* peak ratio (η).

model predicts a dramatic dependence on equivalence ratio (increasing ϕ leads to increased time-zero OH* formation) and a decrease in peak height with increased pressure, yet the model does not reflect the proportional relationship to temperature shown experimentally (depicted previously in Fig. 2b). In an effort to bring model calculations in agreement with experimental results, the measured first-to-second OH* peak ratios (η) were utilized as a tuning parameter for assessing time-dependent kinetics and ignition delay times within the time response limits of the optical setup employed in the experiments.

Prior to replacing rate coefficients for ground-state reactions in the model, the impact of chemiluminescence rates on model predictions of species time histories was determined. The analysis serves to prove that the difference between model and experiment for the OH* time histories is not due in any significant way to uncertainties in the OH* mechanism. The rates of OH* chemiluminescence were varied by multiplying/dividing the associated rate coefficients (by alteration of the frequency factor, A) for the dominant production channels (Eqns. R1 and R2) to assess the level of influence of known OH* chemiluminescence rates on the tuning parameter η . The OH* species rates were used exclusively since the time-zero behavior did not appear in experimental

measurements of CH* time histories. Rates of reactions responsible for producing OH* chemiluminescence were examined to elucidate any impact on the over-predicted OH* concentration near time-zero.



Figure 8 shows the effect of varying A in the rate coefficients for the primary OH* production channels (R1, R2) by factors of 10, where A is the frequency factor in the Arrhenius expression of the rate coefficient $k(T)$, for the conditions: 1000 ppmv n -C₉H₂₀, $\phi = 2.0$, 1555 K, 1.5 atm. The ignition-peak-normalized, model-predicted chemiluminescence profiles of OH* using the initial model [6] were unchanged despite the order-of-magnitude increases or decreases in rate expressions, leading to the conclusion that overprediction of OH* near time zero cannot be attributed to any uncertainty in the chemiluminescence sub-mechanism. The same insensitivity of η to the rates of the OH* formation reactions is observed in both Fig. 8a and Fig. 8b, displayed using a logarithmic vertical scale. Since the chemiluminescence reactions were determined to not affect the tuning parameter η , the experimental OH* time histories were then employed as targets for improving the high-temperature chemistry of the main n -nonane mechanism. To this end, sensitivity, rate-of-production, and reaction pathway analyses were performed at the conditions covered in the experiments to identify ground-state reactions which impact the production/consumption of OH*. Reactions identified as important were then replaced in most cases with literature values or by iterative refinement to (a) improve ignition delay time predictions, (b) improve the OH* profile predictions (i.e. time histories), or (c) determine the mechanism of time-zero OH* formation. R1 and R2 are well-established in the combustion and spectroscopic literature as the primary reactions responsible for the formation of OH*. In general, there are few reactions other than R1 and R2 that are energetic enough to excite OH into the first electronic state. While it is possible that some unknown reactions with hydrocarbon species could also produce electronically excited OH in the n -nonane system, the results shown in Fig. 8 would not change since the rate coefficients were modified by a factor of ten, which covers any OH* formed from unknown reactions.

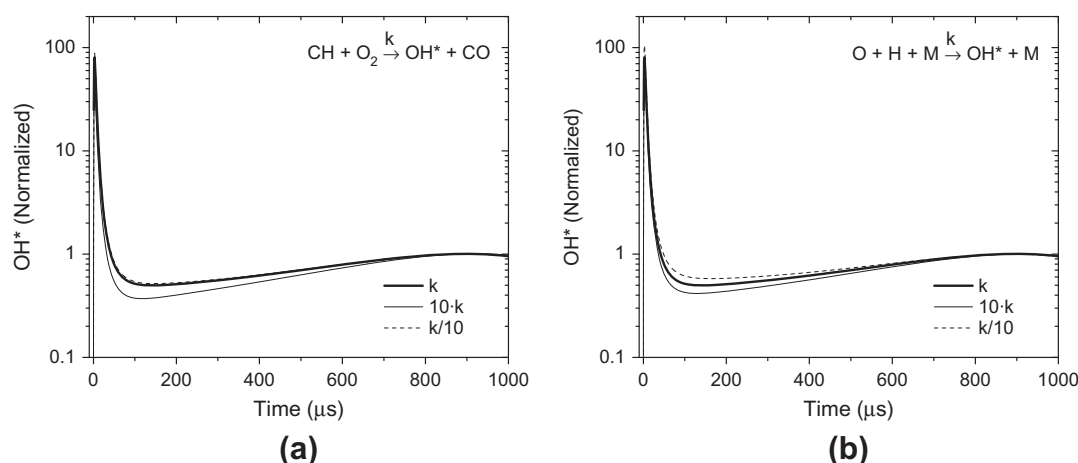


Fig. 8. Calculated OH* mole fraction normalized to the primary ignition peak (near 900 μ s), showing highly overpredicted OH* near time zero. Variation in the rate coefficient of primary reactions producing OH*, (a) R1 and (b) R2, shows no impact on OH* yield near time-zero, indicating an insensitivity of normalized model calculations [6] to chemiluminescence rates. $\eta \gg 1$ in model predictions using [6], in contrast with $\eta < 1$ experimentally (cf. Fig. 7).

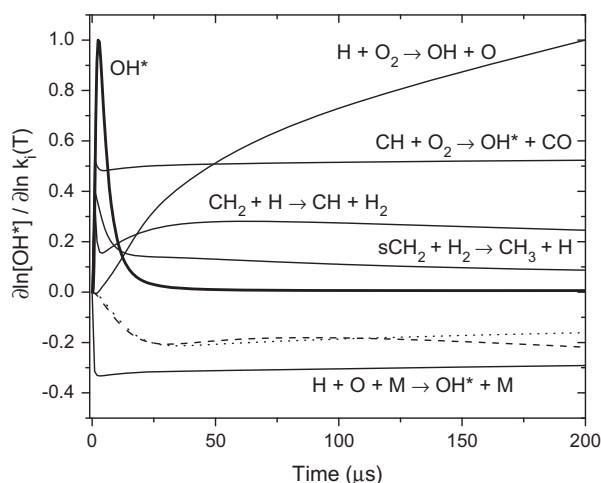


Fig. 9. Representative calculation of normalized OH* mole fraction (bold line) and sensitivity coefficients describing pertinent OH*-sensitive reactions during oxidation steps near time-zero using the initial version of the mechanism [6]; 1555 K, 1.5 atm, $\phi = 2.0$, 99.2% Ar. Dotted line: $C_3H_6 + H \rightarrow aC_3H_5 + H_2$. Dashed line: $C_2H_4 + H \rightarrow C_2H_3 + H_2$.

5. Impact of ground-state reactions on OH* near time zero

Sensitivity coefficients were calculated for OH* with using 1- μ s time steps (Fig. 9) utilizing the SenKin module in ChemKin and employing the standard frequency factor-perturbation approach defined by the equation $S_i = \partial \ln[OH^*] / \partial \ln k_i(T)$. Calculations were performed using three temperatures to cover the extremes for all specific (ϕ , P) conditions. Results from a broad series of sensitivity coefficient computations, determining the ten reactions most sensitive to OH*, were tabulated for all conditions (lean $\rightarrow$ rich, 1.5 $\rightarrow$ 10.5 atm). The results were then cross-compared for identification of overlapping reactions, with dominant reactions analyzed further to assess individual impact on η .

The results for OH*-sensitive coefficients near time zero underscore the dependence of OH* time dependence on ground-state chemistry. Of significance is the insensitivity to time of OH* sensitivity coefficients for the chemiluminescence reactions (Fig. 9), indicating that alteration of the corresponding rates impacts only the absolute magnitude of the OH* profiles, and not the ratio η or profile shape. In contrast, sensitivity coefficients for ground-state reactions involving OH* display a dependence on time near time-zero. Results in Fig. 9 therefore complement the results in Fig. 8: modeled OH* time behavior depends exclusively on ground-state

reaction rates. Calculations carried out at other conditions of equivalence ratio/temperature/pressure were largely similar with respect to the insensitivity of η to the main two OH* reactions, yet involved different ground-state reactions. Prominent, fundamental chain-branching reactions ($H + O_2 \rightarrow OH + O$, etc.) appeared consistently in the sensitivity analysis results, but were not altered since the corresponding rate coefficients are relatively well-known.

Several modifications were implemented into the initial mechanism using results from the sensitivity analyses. Modification of reaction rate parameters (Table 2) include changes in the parent-fuel homolysis reactions (R3–R6); H-abstraction from C_2H_4 by H forming vinyl and H_2 (R7); from CH_2 by H forming methylidyne and H_2 (R9); and by H on formaldehyde yielding formyl and H_2 (R10); and, unimolecular decomposition rates of propene (R8). Initial rate parameters for R7–R10 were taken from the literature. Parameters for R7 were obtained from Weissman and Benson [40], for R8 from Dean [41], for R9 from Frenklach et al. [42], and for R10 from the RRKM extrapolation of Irdam et al. [43]. Homolysis rates for *n*-nonane in the initial version of the model were those used for *n*-decane from the mechanism of Zeppieri et al. [10]. In the present model, *n*-nonane homolysis rates were reduced by decreasing the frequency factor from 2.0×10^{17} to $3.0 \times 10^{16} s^{-1}$. Changes in the rate parameters for several reactions important in the low- and intermediate-temperature regime were applied to the initial version of the model [6] prior to and independent from the present study, including rate parameters for the formation of cyclic ethers from Wijaya et al. [44].

Detailed in Fig. 10 are the specific impacts of individual modifications applied to the initial version of the mechanism. The experimental conditions chosen for comparison were 1000 ppmv *n*-nonane, $\phi = 2.0$, 1477 K, 1.5 atm, due to η being most pronounced at these conditions. The inset of Fig. 10a–c displays the specific impact on the magnitude of the first OH* peak in comparison to predictions from the initial version of the model. Contrast drawn from Fig. 10a–e explicitly highlights the effect of ground-state chemical reactions on the time history of modeled chemiluminescence profiles. The following notation is used to discern amongst the different OH* profiles: 1: experiment; 2: initial model prediction; 3: modified model prediction. Note that results displayed from the series modifications are compounded (i.e. modifications shown in Fig. 10b incorporate those from Fig. 10a, etc). Individual effects of rate modifications on η are outlined in Table 3.

The motivation to improve the previous mechanism centered around three areas: the previous model (1) overpredicted first-to-second peak ratios (η); (2) predicted broader OH* time histories (i.e. broader FWHM relative to experimental measurements); and,

Table 2

Reactions and corresponding rate parameters replaced/modified in [6]. Units: s, cal, mol, cm^3 , K. The initial model expressed R9 in the reverse direction. Methylene (CH_2) in R9 refers to the triplet state (non-zero spin momenta). Initial rate parameters from the previous mechanism in [6] are italicized.

Reaction	A	n	E_a (cal/mol)	Reference
R3: $n-C_9H_{20} \rightarrow pC_4H_9 + AC_5H_{11}$	3.0×10^{16}	0.00	83180.0	Present work
	2.0×10^{17}	0.00	83180.0	[10]
R4: $n-C_9H_{20} \rightarrow nC_3H_7 + AC_6H_{13}$	3.0×10^{16}	0.00	83400.0	Present work
	2.0×10^{17}	0.00	83400.0	[10]
R5: $n-C_9H_{20} \rightarrow C_2H_5 + AC_7H_{15}$	3.0×10^{16}	0.00	83680.0	Present work
	2.0×10^{17}	0.00	83680.0	[10]
R6: $n-C_9H_{20} \rightarrow CH_3 + AC_8H_{17}$	3.0×10^{16}	0.00	85570.0	Present work
	2.0×10^{17}	0.00	85570.0	[10]
R7: $C_2H_4 + H \rightarrow C_2H_3 + H_2$	1.7×10^{13}	0.70	8003.3	[40]
	1.0×10^{14}	0.00	15009.0	[45]
R8: $C_3H_6 \rightarrow C_2H_3 + CH_3$	7.9×10^{16}	0.00	99594.0	[41]
	1.1×10^{21}	–1.20	97720.0	[46]
R9: $CH + H_2 \rightarrow CH_2 + H$	1.2×10^8	1.79	1670.0	[42]
Reverse	3.5×10^{14}	0.00	0	[6]
R10: $CH_2O + H \rightarrow HCO + H_2$	2.9×10^{12}	1.90	2740.4	[43]
	1.1×10^8	1.80	3000.0	[6]

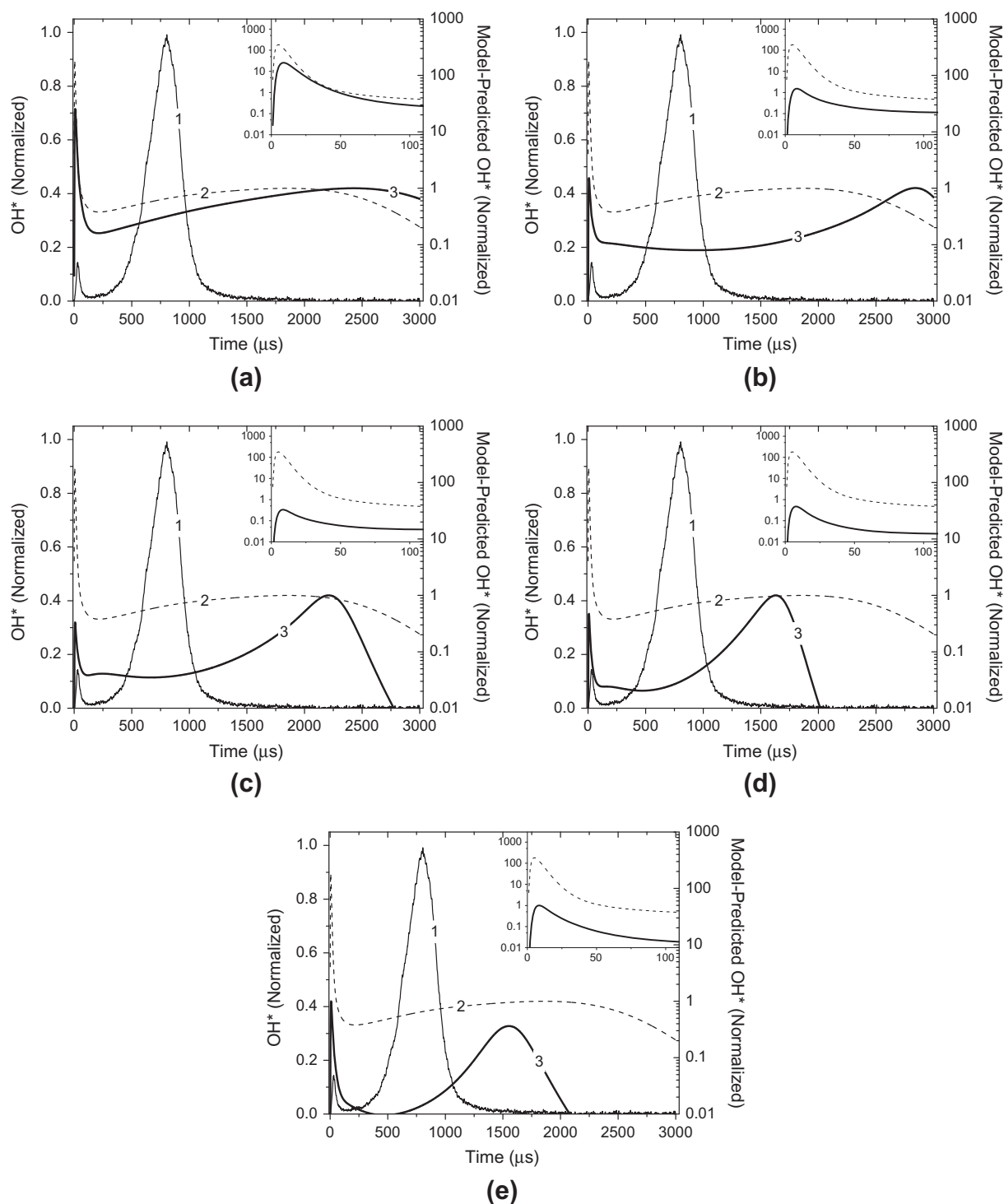


Fig. 10. Impact of rate parameter replacements (Table 2) on modeled OH^* : (a) decreased rate of fuel homolysis reactions, R3–R6; (b) increase in rate of H abstraction from ethylene, forming vinyl + H_2 , R7 (incipient OH^* is reduced by $\sim 10^1$ in both cases, cf. insets); (c) decreased rate of $\text{C}_3\text{H}_6 \rightarrow \text{C}_2\text{H}_3 + \text{CH}_3$, R8; (d) replacement of reaction and rate parameters for $\text{CH} + \text{H}_2 \rightarrow \text{CH}_2 + \text{H}$, R9; (e) increased rate of $\text{CH}_2\text{O} + \text{H} \rightarrow \text{HCO} + \text{H}_2$, R10. The model-predicted OH^* time history is predicted in closer agreement with experiment (i.e. OH^* near time-zero terminates prior to ignition, near 500 μs). Numerical designations 1, 2, and 3 refer to OH^* profiles from experiment, initial model [6], and present model, respectively. Model adjustments are compounded (i.e. rate replacements shown in Fig. 10b incorporate those from Fig. 10a, etc.).

(3) showed an inability to define ignition delay times for fuel-rich conditions. The primary objective of the present work, namely mechanism improvement by mitigation of time-zero OH^* formation through analysis of important ground-state reactions controlling chemiluminescence, is explicitly captured in Figs. 11 and 12. Improvement in areas (2) and (3) are exhibited in Fig. 11, while improvement in area (1) is displayed in Fig. 12. Note in Fig. 11 that

the model-predicted OH^* profiles are peak-shifted such that the shape (i.e. time-dependence) of the two profiles can be unambiguously compared. Although the present model overpredicts ignition delay time at the fuel-rich condition (evident in Fig. 11 by shifting of the OH^* profile to negative times), the effect of modifications applied to the previous mechanism are explicitly shown from the agreeable time-dependent trends between experimental and

Table 3
Summary of modifications (increase/decrease in $k(T)$) incorporated into the present model, relative to respective rate coefficients from Rotavera et al. [6], and corresponding effects on η . The initial model contained Reaction 9 written in reverse with different rate parameters: $\text{CH}_2 + \text{H} \rightarrow \text{CH} + \text{H}_2$, with $A = 3.5 \times 10^{14}$, $n = 0.0$, $E_a = 0.0$ kcal/mol.

Reaction	$k(T)$	Impact
R3: $n\text{-C}_9\text{H}_{20} \rightarrow \text{pC}_4\text{H}_9 + \text{AC}_5\text{H}_{11}$	Decreased	Decreased η by ~ 10
R4: $n\text{-C}_9\text{H}_{20} \rightarrow \text{C}_3\text{H}_7 + \text{AC}_6\text{H}_{13}$	Decreased	
R5: $n\text{-C}_9\text{H}_{20} \rightarrow \text{C}_2\text{H}_5 + \text{AC}_7\text{H}_{15}$	Decreased	
R6: $n\text{-C}_9\text{H}_{20} \rightarrow \text{CH}_3 + \text{AC}_8\text{H}_{17}$	Decreased	
R7: $\text{C}_2\text{H}_4 + \text{H} \rightarrow \text{C}_2\text{H}_3 + \text{H}_2$	Increased	Decreased η further by $\sim 10^1$
R8: $\text{C}_3\text{H}_6 \rightarrow \text{C}_2\text{H}_3 + \text{CH}_3$	Decreased	(1) Improved OH^* profile shape (2) Decreased η further by $\sim 10^1$ (3) Definable ignition at $\phi = 2.0$
R9: $\text{CH} + \text{H}_2 \rightarrow \text{CH}_2 + \text{H}$	–	(1) Improved OH^* profile shape (2) Improved ignition times
R10: $\text{CH}_2\text{O} + \text{H} \rightarrow \text{HCO} + \text{H}_2$	Increased	Improved OH^* profile shape

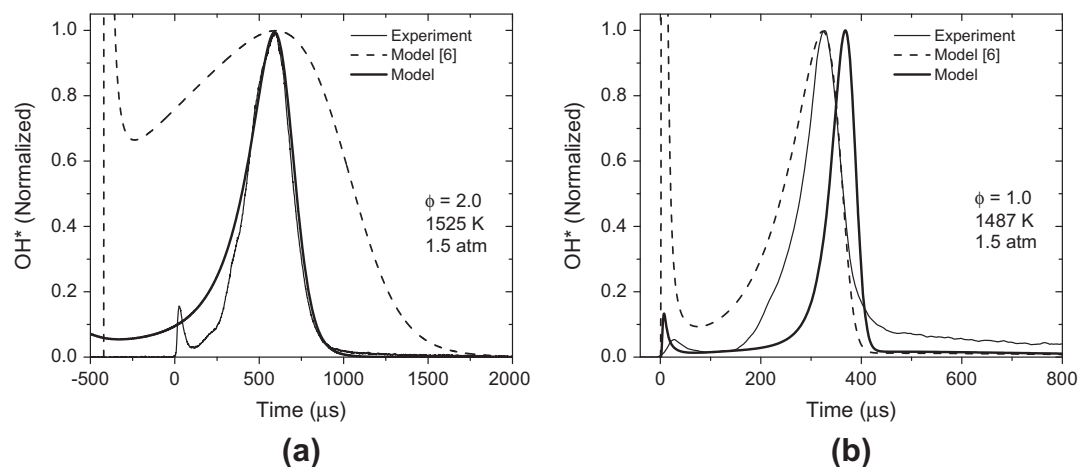


Fig. 11. Improved model predictions of OH^* time histories. Replacement of rate parameters of ground-state reactions (R3–R10, Table 2) led to increased accuracy in predictions of OH^* time histories. Similar results were obtained at other conditions. (a) peak-shifted profile showing improved prediction of OH^* time dependence at $\phi = 2.0$; ignition delay times, previously indefinable, are definable using the present mechanism; (b) representative OH^* profile showing improved prediction of time-zero OH^* .

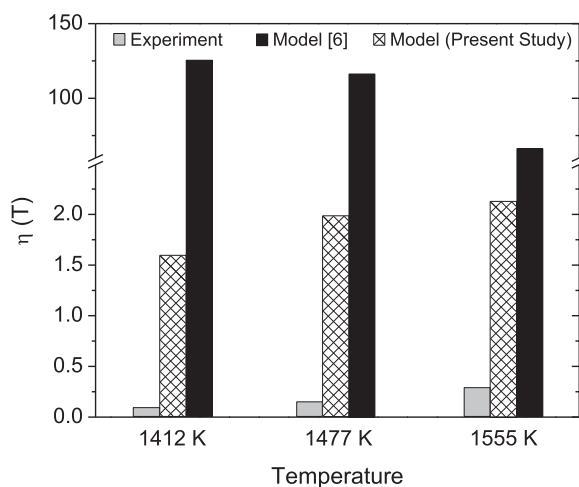


Fig. 12. Improved temperature dependence of $\eta(T)$ resulted from replacement of rate parameters for reactions R1–R9 (Table 2), identified by OH^* sensitivity analysis and related effects on OH^* time histories. Conditions: 1000 ppmv $n\text{-C}_9\text{H}_{20}$, $\phi = 2.0$, 1.5 atm. The present model exhibits a significantly decreased η and corrected temperature trend, as observed experimentally. The precise magnitude of η depends on the PMT output resistance (cf. Fig. 4).

model-predicted profiles. Figure 12 depicts a representative comparison of the improvements with respect to η . Under the conditions shown for Fig. 12, at 1477 K, the previous model [6] predicted $\eta = 117$, while the improved model (unfiltered for time response) predicts $\eta = 2.0$ (experimentally: $\eta = 0.16$). The slight discrepancy in η between the experimental result and that of the

present model is acceptable given the dependence of PMT output resistance on quantifying η . In fact, the residual over-prediction of η by the present kinetics model for n -nonane is closer to the correct result since the time response of the detector setup led to first peaks that could be as much as a factor of two lower (cf. Fig. 4). Some further discussion with respect to correcting the predicted time histories to match the measurement limitations is provided in subsequent sections, where the current model, when filtered for the time response of the optical system, matches the measured results quite well in terms of the timing and magnitude of the first OH^* peak. Of higher importance is that η is less than unity, whereas previous predictions indicated significantly larger values. Moreover, the modifications applied to the initial mechanism resulted in the $\eta(T)$ trend coming into agreement with experimental results. Specifically, $\eta(T)$ increases with increasing temperature (Fig. 12).

6. Mechanism of time-zero OH^* formation

To characterize the chemical mechanism of OH^* formation near time-zero, reaction pathway analysis centered around the formation of methylidyne (CH), given the dominant OH^* production channel: $\text{CH} + \text{O}_2 \rightarrow \text{OH}^* + \text{CO}$ (R1). Formation of CH stems primarily from a series of H-abstraction reactions commencing with the formation of methyl (CH_3), with methyl being produced from a range of reactions including unimolecular decomposition of n -nonane. The significance of methyl arises from the yield of singlet-state methylene (sCH_2) coming solely from $\text{CH}_3 + \text{H} \rightarrow \text{sCH}_2 + \text{H}_2$. De-excitation of singlet-state methylene to CH_2 then leads via H-abstraction to CH . Although side reactions varied with different conditions, the general pathway to CH remained consistent in that

results were similar for all equivalence ratios considered. With increasing time during the first 5–10 μs , radical–radical reactions led to methyl consumption and formation of alkene intermediates, therein reducing the amount of CH present for OH^* production.

Using the present chemical kinetics mechanism, reaction pathway analyses (RPA) on dilute (99.2% Ar) oxidation of *n*-nonane under fuel-rich conditions ($\phi = 2.0$) were performed at an arbitrary high temperature of 1525 K and 1.5-atm pressure to delineate the mechanism of OH^* formation near time-zero at three discrete times: 3, 5, and 15 μs (Fig. 13). The analysis, conducted on 0.1- μs intervals over the first 50 μs of reaction, shows that initial fuel decomposition leads through C–C bond fission immediately to alkyl radicals which then proceed through a series of elementary reactions, ultimately leading to the formation of CH through $\text{CH}_2 + \text{H} \rightarrow \text{CH} + \text{H}_2$. Production of CH by H-abstraction from methylene remained the dominant pathway for all equivalence ratios and pressures considered. The formation of alkyl radicals from the parent fuel results from the high temperatures in the experiments and calculations (>1200 K), wherein fuel decomposition pathways compete strongly with H-abstraction reactions leading to alkyl radicals being produced largely through unimolecular decomposition ($n\text{-C}_9\text{H}_{20} \rightarrow \text{R}^\cdot + \text{R}'^\cdot$).

The general reaction mechanism under fuel-rich conditions is illustrated in Fig. 14. Results under lean and stoichiometric conditions were largely similar. Model calculations using the present mechanism predict the following with respect to unimolecular fuel decomposition: the decomposition channel yielding 1-butyl + 1-pentyl consumes 31% of *n*-nonane, while channels forming *n*-propyl + 1-hexyl and ethyl + 1-heptyl consume 29% and 26% of *n*-nonane, respectively, and the least-preferred decomposition channel yields methyl + 1-octyl (14% of *n*-nonane consumption). After the initial fuel decomposition step, the fragmented fuel components (alkyl radicals) react in a series of steps which lead to the formation of CH_3 , after which the pathway to OH^* remains independent of time. Taken collectively, Fig. 14 outlines the formation mechanism of OH^* near time-zero. The calculations indicate rapid unimolecular decomposition, first of *n*-nonane, then of *n*-propyl and *n*-pentyl radicals yielded from β -scission, leads to methyl radicals which then follow the pathway in Fig. 14. With increasing time, towards the peak of OH^* near 5 μs , the displacement reaction

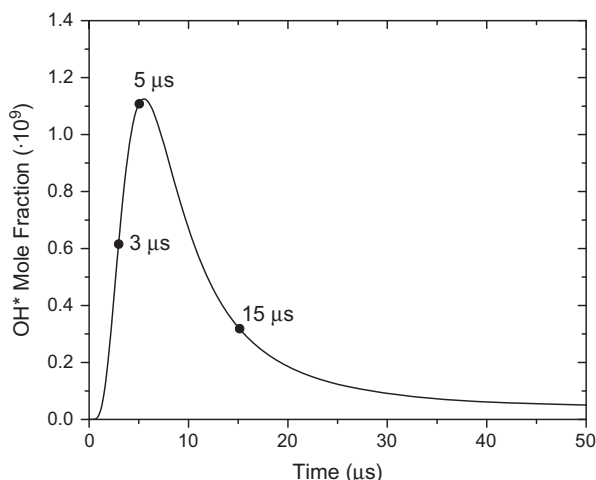


Fig. 13. Discrete times selected for reaction pathway analysis of OH^* formation near time-zero. The three points at 3-, 5-, and 15- μs represent, respectively, initial OH^* formation, change in the rate of OH^* formation, and diminished rate of OH^* formation. The overall mechanism (Fig. 14) remained the same, yet reaction rates of the involved species varied with time. Increasing rates of CH_2 depletion from separate reaction with CH_3 and O_2 , starting at times near 5- μs , instigate the rate of OH^* depletion at times post-peak yield.

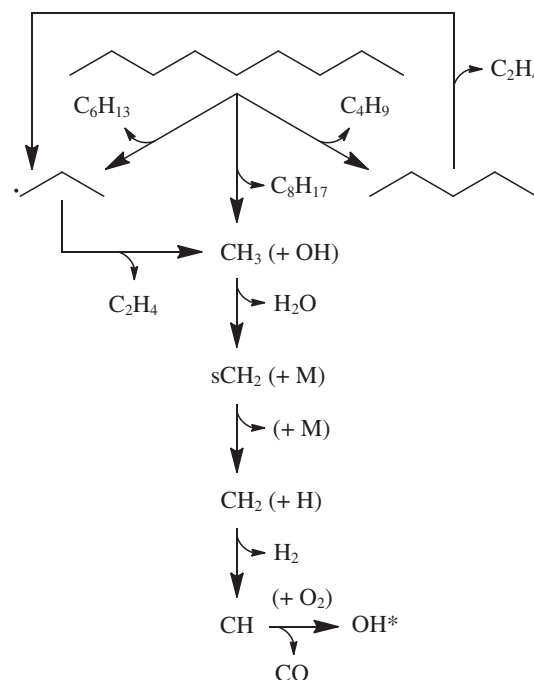


Fig. 14. Dominant reaction pathway of initial OH^* formation; $\phi = 2.0$, 1525 K, 1.5 atm, 1000 ppmv *n*-C₉H₂₀. The rate of OH^* formation near time-zero decreases with time due to depletion of CH_2 by CH_3 and O_2 , forming $\text{C}_2\text{H}_4 + \text{H}$ and $\text{CO}_2 + \text{H}_2$, respectively. sCH_2 refers to singlet-state methylene. The ethyl + *n*-heptyl pathway is not predicted to appreciably yield methyl directly (i.e. similar to *n*-propyl or from *n*-nonane) or indirectly (i.e. *n*-pentyl $\rightarrow$ *n*-propyl $\rightarrow$ methyl).

of methylene with methyl forming $\text{C}_2\text{H}_4 + \text{H}$ in addition to $\text{CH}_2 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}$ contributes to an overall lower rate of CH production and, ultimately, diminishing rate of OH^* . The diversion of CH_2 therefore instigates the slowing of time-zero OH^* formation since CH_2 directly yields CH from H-abstraction. In general, the mechanism of OH^* near time-zero in *n*-nonane oxidation (likely similar to other higher-order *n*-alkanes) can be generalized into six steps:

- **Step 1:** Homolysis of *n*-nonane (formation of 1-pentyl, 1-butyl, 1-propyl, ethyl, methyl radicals).
- **Step 2:** Formation of methyl (from alkyl β -scission and fuel homolysis).
- **Step 3:** Methyl radicals react with H or OH producing singlet-state methylene (sCH_2).
- **Step 4:** Collisional de-excitation of singlet-state methylene forms methylene (CH_2).
- **Step 5:** H-abstraction from methylene by H produces CH.
- **Step 6:** CH reacts with O_2 to form $\text{OH}^* + \text{CO}$.

7. Characterization of OH^* chemical kinetics near time-zero

For all equivalence ratios, η strongly increased with increasing temperature. To illustrate, Fig. 15 shows a juxtaposition of two OH^* measurements taken under fuel-rich conditions at 1412 K and 1555 K in 99.2% Ar near 1.5 atm. The 140-K increase in temperature led to an increase in first-peak OH^* formation relative to the second peak OH^* by a factor of 3.5. While the chemistry differs between the formation of OH and OH^* , this observed phenomenon mirrors the oxidation behavior observed near time-zero in the ground-state hydroxyl measurements of Vasu et al. [47] during *n*-dodecane oxidation. The laser absorption measurements in [47] indicated OH formation immediately behind the reflected shock wave (time-zero), marked by an appreciable increase in OH absorption in both fuel-lean and stoichiometric mixtures. Similar

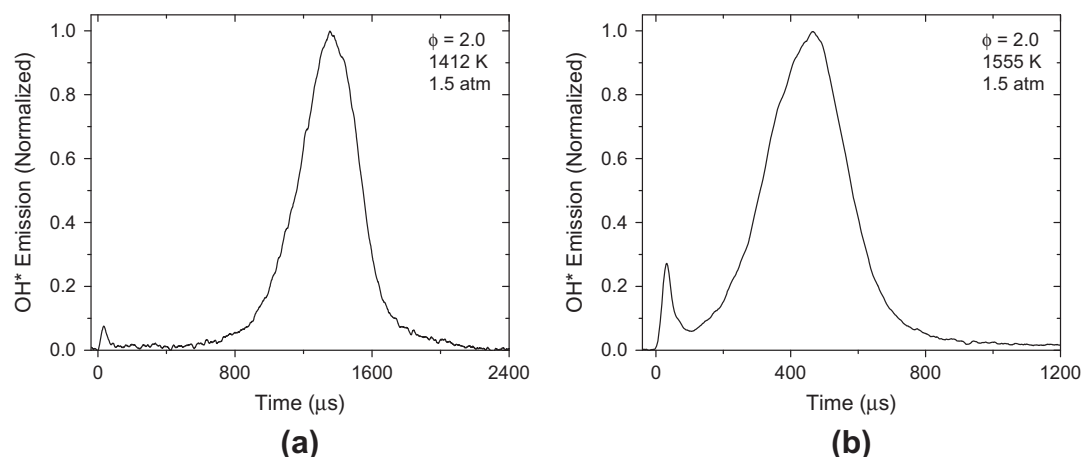


Fig. 15. Increased temperature leads to higher rates of OH* formation near time-zero at fixed equivalence ratio; (a) 1412 K ($\eta = 0.08$) and (b) 1555 K ($\eta = 0.27$).

to the present work, the work of Vasu et al. utilized highly dilute conditions, with an initial *n*-dodecane concentration of 1000 ppmv. Near 1420 K and 15.5 atm, a 20-ppmv level of OH was measured and sustained for approximately 20 μ s. The initial oxidation steps leading OH near time-zero were not detected at lower temperatures. The temperature dependence is also reflected in the OH* measurements in the present study. The Vasu et al. study employed four chemical kinetics models, Ranzi et al. [48], Montgomery et al. [49], Zhang et al. [50], and You et al. [51] to compare with the OH concentration measurements, and predictions of the OH plateau near time-zero were 16, 18, 10, and 57 ppmv, respectively.

7.1. Dependence of η on equivalence ratio

For fixed conditions of temperature and fuel concentration, increasing equivalence ratio also causes increased growth of the first OH* peaks. Comparison of three OH* measurements at 1478 K near 1.5 atm shows the dependence of post-shock OH* on equivalence ratio for a fixed fuel concentration of 1000 ppmv *n*-nonane (Fig. 16a). The inset in Fig. 16a details the relative behavior for the duration of the first peaks. To illustrate the broader effect of equivalence ratio on η under conditions of fixed fuel concentration, Fig. 16b delineates the temperature dependence of η for $\phi = 0.5$, 1.0, and 2.0 from the oxidation of 1000 ppm of *n*-nonane. The experimental trends indicate that η is strongly dependent on equivalence ratio for a given temperature. From calculations using the present model, the effect of equivalence ratio on η is due to a higher concentration of CH produced during initial fuel decomposition, increasing by a factor of nearly three with increasing equivalence ratio from 0.5 to 2.0 at both 1.5 atm and 10.0 atm. For all conditions, H-abstraction from ground-state methylene yielding CH and H₂ proved to be the dominant source of CH. Methylidyne depletion in the first 10 μ s comes predominantly from $\text{CH} + \text{O}_2 \rightarrow \text{CO} + \text{OH}$, with secondary and tertiary consumption from $\text{CH} + \text{O}_2 \rightarrow \text{CO} + \text{OH}^*$ and $\text{CH} + \text{CH}_3 \rightarrow \text{C}_2\text{H}_3 + \text{H}$, respectively.

Using the present mechanism, model calculations were performed at constant initial fuel concentration (1000 ppmv) over $0 < t (\mu\text{s}) < 20$ to determine the governing reactions leading to the observed higher OH* peak ratio under fuel-rich conditions ($\phi = 2.0$) compared to fuel-lean ($\phi = 0.5$) conditions. The temperature and pressure conditions chosen for the calculations were 1500 K and 1.5 atm, respectively. Calculations were repeated at 10 atm. Noting that the dominant source of OH* is R1 ($\text{CH} + \text{O}_2 \rightarrow \text{OH}^* + \text{CO}$), the analysis focused exclusively on methylidyne and molecular oxygen. The results from calculations of species concentration and rate-of-production were sufficient to identify the

underlying chemical reactions governing the dependence of OH* on equivalence ratio.

Within the first 20 μ s of oxidation, the rate of CH yield is shown to be higher for the fuel-rich equivalence ratio by a factor of approximately two relative to the CH yield under fuel-lean conditions (Fig. 17a). Since initial fuel concentration is held constant in the calculations at 1000 ppmv, the higher rate of CH formation in the fuel-rich case led to the yield of CH being nearly twice the yield in the fuel-lean case over the first 20 μ s. Results in Fig. 17a for 1.5 atm were consistent at 10-atm pressure. From the increased CH present at the fuel-rich condition, analysis then focused on differences in the mechanism of oxygen consumption as a function of equivalence ratio. The rates of consumption of O₂ by $\text{CH} + \text{O}_2 \rightarrow \text{OH}^* + \text{CO}$ (R1) relative to the total consumption rate by all reactions in the model were calculated for both equivalence ratios (Fig. 17b). The level of analysis displayed in Fig. 17 indicates that under fuel-rich conditions for fixed initial fuel concentration, CH is formed at a higher rate, and the rate of oxygen consumption by R1 is also higher, a coupled effect leading to increased OH* near time-zero (following the pathway delineated in Fig. 14).

It is conceivable that, although the fuel-lean equivalence ratio yields less CH relative to the fuel-rich equivalence ratio, the increased O₂ present in the lean condition provides excess oxygen to R1 and therefore leads to an increased rate of OH* yield via R1. However, calculations of the five most-dominant oxygen-consumption reactions (cumulatively representing 94% of total oxygen consumption in the first 20- μ s) using the present mechanism indicate that, while more O₂ is present at the lean condition, the excess O₂ is not consumed by CH. Rather, O₂ is more strongly consumed by other radicals, particularly vinyl (C₂H₃), formyl (HCO) and atomic hydrogen for chain-branching (Fig. 18). A notable difference in the two cases of equivalence ratio is that the consumption of O₂ by HCO increases substantially (and disproportionately) compared to the other four dominant consumption reactions. The net results from comparing Fig. 18a and 18b support the results depicted in Fig. 17 with respect to the equivalence ratio-dependence of OH*.

The above analysis makes clear the increase in η with equivalence ratio for conditions of fixed initial fuel concentration. Specifically, the coupled effect of increased CH yield and the higher rate of O₂ consumption specifically by R1 at $\phi = 2.0$ are responsible for the phenomenon. In contrast, η is observed to be lower for fuel-lean conditions. The reduction in η under lean conditions is due largely to the significantly higher rate of O₂ consumption by reactions other than R1 despite the higher initial oxygen concentration (from Rotavera et al. [6], oxygen percentages for $\phi = 2.0$ and $\phi = 0.5$ were 0.7% and 2.8%, respectively).

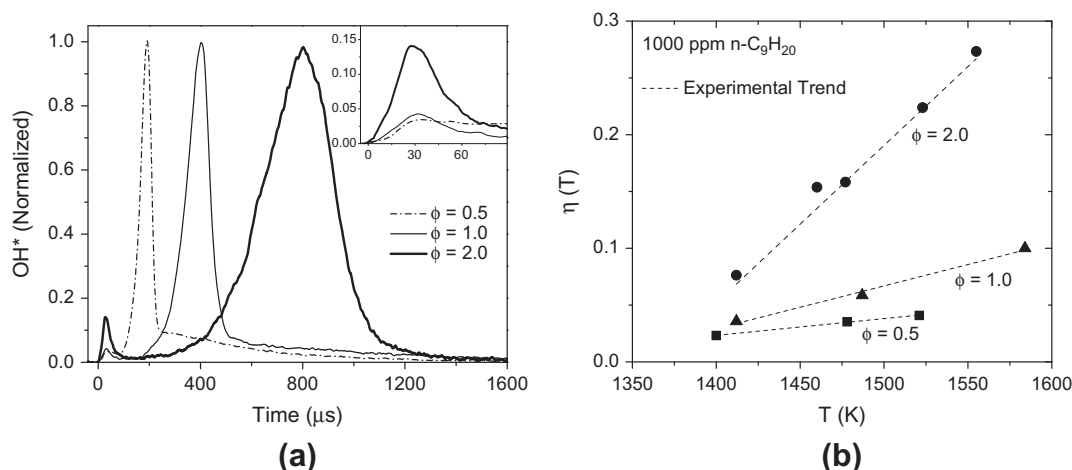


Fig. 16. (a) Experimental OH* profiles near 1478 K near 1.5 atm at fixed fuel concentration of 1000 ppmv showing the dependence of η on stoichiometry. Relative first-to-second peak ratios were quantified: $\phi = 0.5$ ($\eta = 0.035$), $\phi = 1.0$ ($\eta = 0.045$), $\phi = 2.0$ ($\eta = 0.150$). (b) Experimental trends of η produced from oxidation of 1000 ppm of *n*-nonane at 1.5 atm. First-peak OH* relative to peak OH* post-ignition shows a strong dependence on equivalence ratio for conditions of constant fuel concentration. PMT resistance: 4.8-k Ω .

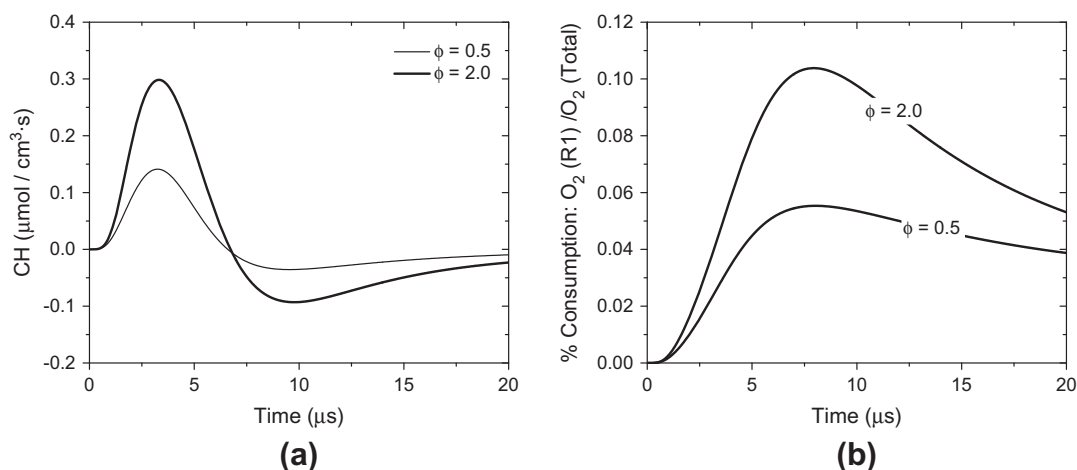


Fig. 17. Fuel-lean and fuel-rich calculations at 1500 K and 1.5 atm near time-zero of (a) CH production and (b) consumption of O₂ by CH + O₂ → OH* + CO (R1) relative to the overall rate of O₂ consumption using the present mechanism; initial fuel concentration: 1000 ppmv *n*-C₉H₂₀. Similar results were calculated at 10-atm pressure, with higher yield at the lean condition and the overall process terminating near 5 μs.

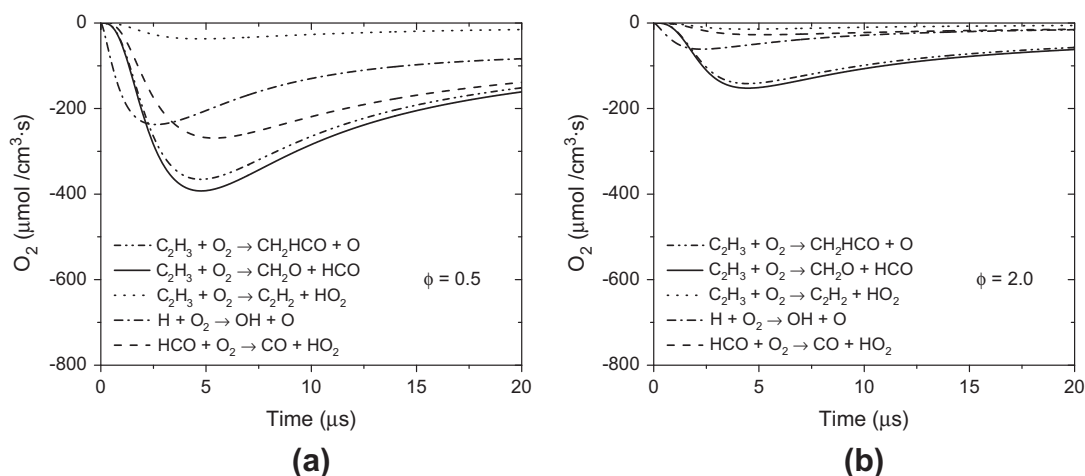


Fig. 18. Rate of O₂ consumption near time zero by the five most-dominant reactions, cumulatively representing 94% of total oxygen consumption for (a) fuel-lean and (b) fuel-rich conditions at 1500 K, 1.5 atm, 1000 ppmv *n*-C₉H₂₀. Despite the higher concentration of O₂ under lean conditions, oxygen is consumed more rapidly by reactions other than the dominant OH* formation channel (R1). Reaction of O₂ with formyl (HCO) assumes a more-dominant role under fuel-lean conditions, contributing in part to the overall higher rate of oxygen consumption at $\phi = 0.5$.

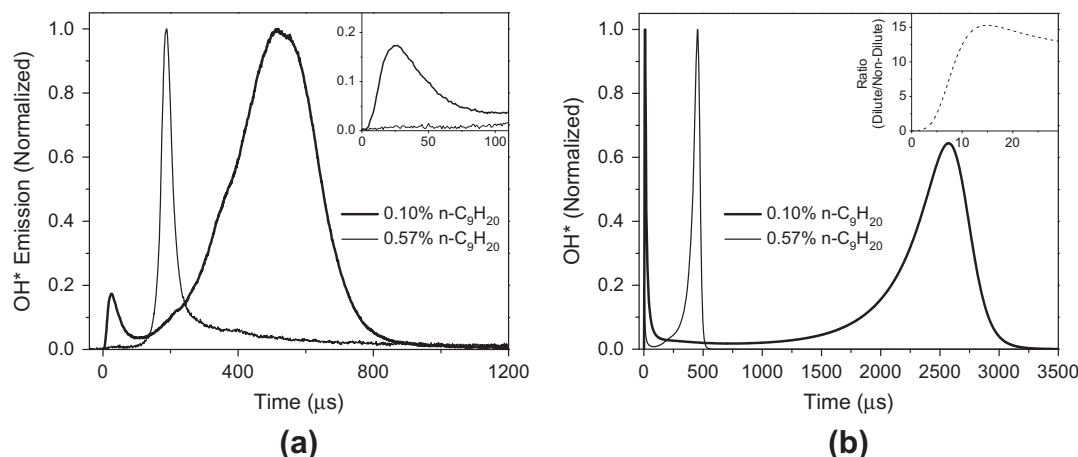


Fig. 19. (a) Effect of increased fuel concentration on experimental formation of OH* at fixed equivalence ratio ($\phi = 2.0$) and temperature (1515 K) near 1.5 atm. At 99.2% and 95.4% Ar dilution, the fuel concentration is 0.10% and 0.57% by volume, respectively. (b) Model calculations of OH* using the present mechanism under dilute and non-dilute conditions showing the impact of fuel concentration on first-peak OH*. Inset of (b) shows that OH* is produced in higher quantities at higher levels of dilution.

7.2. Dependence of η on initial fuel concentration

The effect of fuel concentration on η , which is coupled to a change in diluent concentration for the same equivalence ratio, was investigated by selecting conditions where the time-zero phenomenon was most pronounced: fuel-rich, high-temperature, low-pressure conditions. The effect of n -nonane concentration on OH* at the first peak was determined using $\phi = 2.0$ experiments conducted at 1.5 atm near 1515 K in 95.4% Ar and 99.2% Ar using a 10-k Ω PMT resistance (Fig. 19). In contrast to comparison of traces under conditions of similar dilution, the more-dilute profile displays a larger η compared to the less-dilute profile (Fig. 19a). Model predictions of OH* profiles exhibit similar results (Fig. 19b). One plausible explanation for the smaller amount of behavior near time-zero in the profile measured in the higher-concentration system could be the higher concentration of OH* formed post-ignition and consequently overshadowing of the smaller, profile near time-zero.

8. Ignition delay time modeling

Ignition delay time predictions were also improved as a result of the analyses and subsequent rate parameter replacements.

Figs. 20 and 21 compare experimental results to model predictions from the previous mechanism [6] and the present mechanism. Two sets of experimental results are utilized for comparison, one in which oxygen remained fixed at 4% and one in which n -nonane remained fixed at 1000 ppmv. Stoichiometric results obtained by the authors at 4% oxygen were reported in Rotavera and Petersen [1]. Lean and stoichiometric results from Davidson et al. [3] (at 4% oxygen) are reported for comparison. The fuel-lean ($\phi = 0.5$) and fuel-rich ($\phi = 2.0$) results in Figs. 20 and 21, respectively, were obtained as part of the present study. The 1000-ppmv n -nonane results are reproduced from Rotavera et al. [6]. Experimental results are compared to model calculations of ignition delay times using JetSurF v2.0 [52] and the n -alkane mechanism from Westbrook et al. [13], and are reported only for the 4% oxygen condition and not for the other condition to maintain clarity since measurements are reported for the former at only one pressure (1.5 atm).

The disparity in predictions amongst the four different models can, in general, be attributed to differences in the rate parameters of certain fundamental reactions [1]. Of importance to the present work is the overall improvement in ignition delay time predictions of the present model relative to the initial version [6], notably in Figs. 20 and 21. Moreover, the present model allows for clear definition of ignition delay time at fuel-rich conditions (Fig. 21d),

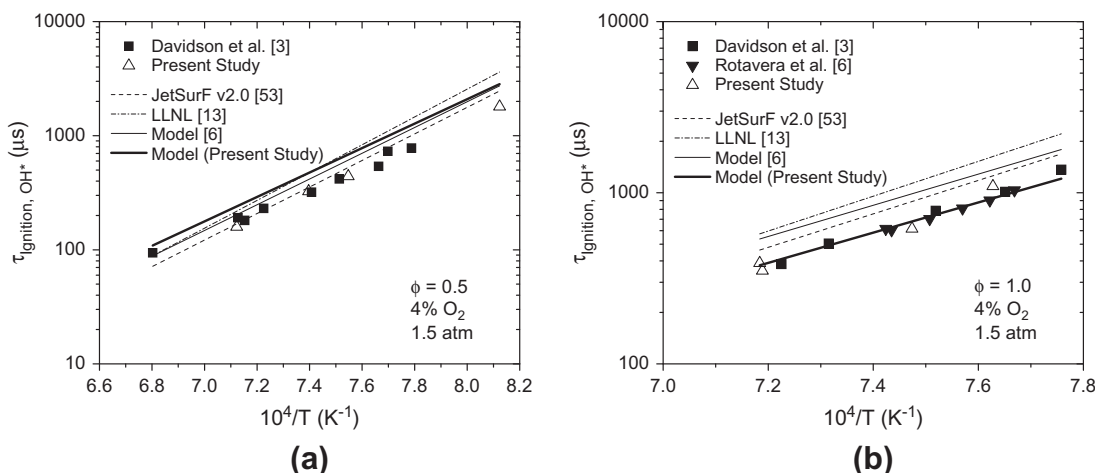


Fig. 20. (a) Fuel-lean and (b) stoichiometric ignition delay time measurements and model predictions of n -nonane in 4% O₂ near 1.5 atm compared with measurements from Davidson et al. [3]. Replacement of rate parameters for R3–R10 led to improved ignition delay time predictions relative to the initial model [6].

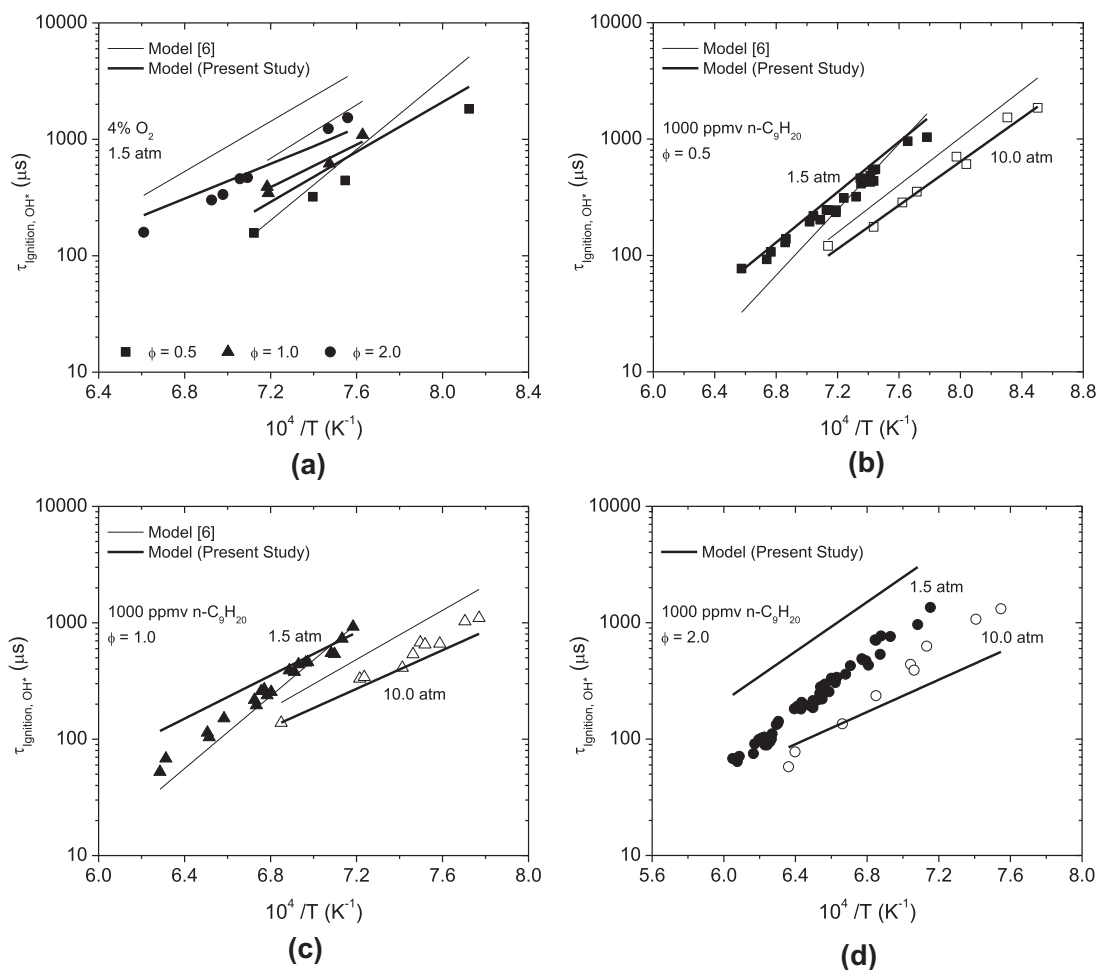


Fig. 21. Ignition delay time measurements of *n*-nonane in O_2/Ar and comparison to model predictions using the initial model [6] and the present model. (a) Fixed oxygen concentration (4% vol.); (b)–(d) 1000 ppmv *n*-nonane, $\phi = 0.5, 1.0, 2.0$, respectively. Replacement of rate parameters for R3–R10 led to improved ignition delay time predictions relative to the initial model [6] and to definable OH^* -based ignition delay time predictions at $\phi = 2.0$. OH^* -based ignition delay times were indefinable using the first version of the mechanism [6] due to strong overprediction of time-zero oxidation processes yielding OH^* that resulted in unclear ignition definition.

although overprediction at lower pressure is still apparent by a factor of approximately three. Overall ignition activation energy at $\phi = 2.0$ is captured well at 1.5 atm, yet sharply underpredicted at 10.3 atm by approximately 31.6 kcal/mol (compared to 52.4 kcal/mol experimentally) indicating that the pressure dependence at the fuel-rich condition requires improvement.

9. Comparison of OH^* and CH^* near time-zero

Processes similar to those leading to OH^* formation in the first 20 μs of *n*-nonane decomposition were not observed in CH^* measurements under any conditions. Noting the significant dependence of time-zero processes on temperature for OH^* , experiments were conducted at higher temperatures for measurement of CH^* time histories in an attempt to magnify any underlying processes. Measurements of CH^* time histories did not display the two-peak behavior characteristic of OH^* , even under comparatively higher temperatures (Fig. 22). The temperature difference between the OH^* and CH^* profiles of 80 K in Fig. 22 does not reveal an initial peak near time-zero for CH^* , despite the high temperature.

Rate-of-production simulations on CH^* near time-zero indicate that production and consumption rates are effectively equivalent, resulting in negligible CH^* production on the order of 10^{-9} ppm

(Fig. 23); similar calculations indicated a net OH^* production on the order of 10^{-2} ppm under the same conditions. Results were similar for other equivalence ratios and temperatures and were independent of pressure. Although the reactions responsible for CH^* production differ from those for OH^* , it is noteworthy that CH^* is strongly quenched by third-body collisions (principally with Ar), while under the same conditions OH^* quenching is unaffected by collisional de-excitation relative to production rates. From similar analysis, the main consumption channels of OH^* near time-zero were $OH^* + Ar \rightarrow OH + Ar$ and $OH^* + O_2 \rightarrow OH + O_2$; however, the collision rates were more than ten times slower than the net production rate of OH^* .

10. Filtered response of improved kinetics model

As shown previously when discussing the time response of the experimental optical setup for detecting OH^* , a simple model was formulated that can reproduce the effect of the experiment time response on the measured OH^* time history, namely during the first peak near time zero (cf. Fig. 6). The simple RC model is used to process the numerical OH^* predictions from the kinetics model to replicate the time-response of the experiment. Such a calculation is shown in Fig. 24, using the data and predictions from Fig. 11b. In Fig. 24a, both the initial model and the improved model

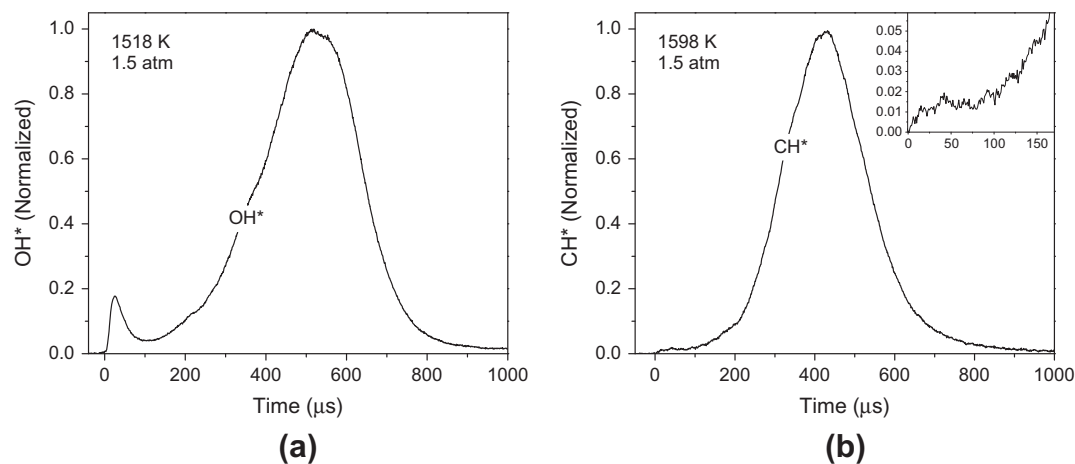


Fig. 22. Comparison of excited-state species time profiles during *n*-nonane oxidation. (a) OH^{*}; (b) CH^{*}. Despite the nearly 80-K higher temperature of the CH^{*} profile, no significant peak near time-zero is evident.

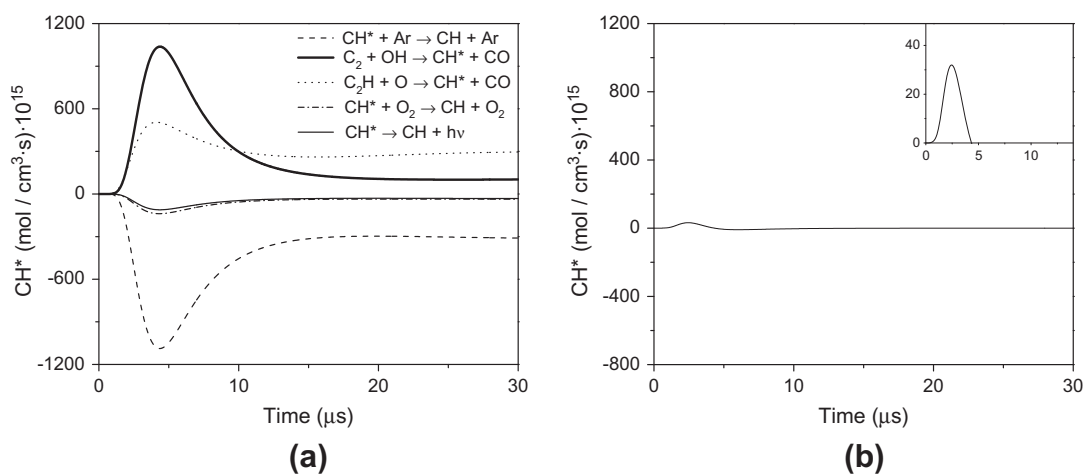


Fig. 23. (a) Rates of CH^{*} production/consumption and corresponding dominant reactions calculated using the present model. (b) Net rate of CH^{*} production. Since production of CH^{*} closely equals consumption within the first 10 μs, no CH^{*} is predicted, supporting experimental observations. 1000 ppmv *n*-C₉H₂₀; $\phi = 2.0$; 1598 K; 1.5 atm.

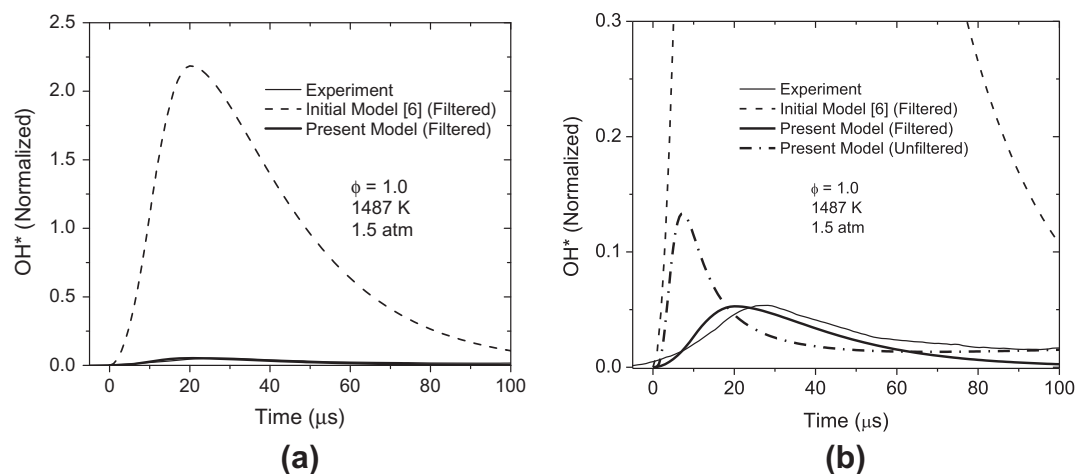


Fig. 24. Results of the chemical kinetics model processed by the RC-circuit filter for the first OH^{*} peak, in comparison with the experiment for the results initially shown in Fig. 11b (1487 K, 1.5 atm, $\phi = 1.0$); (a) comparison showing both the initial [6] and present models; (b) comparison emphasizing the current model – filtered and unfiltered – in relation to the experiment.

of the present study are shown in comparison with the experiment (1487 K, 1.5 atm, $\phi = 1.0$); the kinetics model calculations in Fig. 24 were processed using the RC filter model. As shown in Fig. 24a, the initial model prediction greatly over-predicts the first OH* peak. Even when processed by the RC filter model, the calculated profile is far removed from the experimental result. This calculation confirms the validity of the experiments herein for tailoring the chemical kinetics model. That is, if the actual OH* first peak were as large as that predicted by the initial model, the measured time history would resemble the dashed line in Fig. 24a.

Figure 24b shows a close-up of the region near time-zero to emphasize the filtered, improved kinetics model and its comparison with the experimental data. Remarkably, the simple RC-circuit model in conjunction with the updated kinetics model matches the experimentally observed first peak in magnitude and timing extremely well. The processed calculation using the improved model also matches the experiment much closely than the unfiltered calculation, which assumes the response time of the measurement system is zero. Figure 24b further validates the modifications made to the model herein via the one-to-one comparison between the kinetics model and the actual experimental results. Similar results can be shown for the other conditions examined in this study. Note that only the first peak is modeled in Fig. 24, thus the filtered calculation tends toward zero at the later times, while the experimental OH* profile starts to merge with the main ignition peak at later times.

11. Concluding remarks

OH* time histories obtained in recent shock-tube measurements were used to better understand the kinetics of *n*-nonane oxidation at incipient times. OH* profiles were particularly useful for tuning the relative kinetics due to a unique, two-peak behavior between the initial oxidation steps of the parent fuel molecule and primary ignition. Improvements in the predictive capability of a chemical kinetics model for *n*-nonane oxidation were implemented, leading to more-accurate ignition delay time and species time history predictions at high temperatures using the ratio between the two observed peaks in measured OH* profiles as a model-performance metric. Chemical kinetics leading to OH* immediately after shock passage are governed by the production of CH radicals which are produced by a number of ground-state reactions that ultimately supply CH to the dominant OH* production channel: $\text{CH} + \text{O}_2 \rightarrow \text{OH}^* + \text{CO}$. Although side reactions differed from condition to condition, reaction pathway calculations revealed the general mechanism of time-zero OH* formation to be largely similar for all conditions studied, with a range of reactions leading to CH₃ followed strictly by a series of H-abstraction reactions (and de-excitation of single-state methylene) leading to CH.

In the present work, modifications to ground-state reactions revealed using OH* sensitivity analysis were implemented primarily by replacing rate parameters existing in the initial version of the model with parameters from the literature. Fuel homolysis reaction rates were reduced by lowering the frequency factor, *A*, in the Arrhenius equations. However, an alternative approach could also lead to improvements, including analysis of activation-energy effects on key processes, thermochemistry, and thermodynamics. Analysis centered on improving model predictions of OH* led to four primary results: (1) accurate prediction of early time oxidation processes near time-zero (quantified by the ratio of the first-to-second OH* peak, η); (2) improved prediction of OH* time-dependence/profile shape; (3) definable OH*-based ignition delay times for $\phi = 2.0$; and, (4) improved ignition delay time predictions. Further improvement in model accuracy at fuel-rich conditions is still needed, yet in the context of the present work, model predictions of fuel-rich ignition delay times became definable using OH*

measurements from changing certain ground-state reaction rate parameters. Ignition delay times were initially indefinable using the initial mechanism due to unclear ignition events from overpredicted OH* peaks near time zero. Also of significance herein is the development of a simple RC-circuit model of the optical detection system through which the kinetics predictions could be processed. The end result is a method that allows the predictions to model the actual conditions of the experiment, taking into account time-response, rather than an idealized version. Such a procedure should serve as a methodology for comparing such OH* (and other) time histories between prediction and measurement.

While studying the time dependence of excited-state species has utility as described herein, the authors do not suggest that chemiluminescence time histories uniquely satisfy the multitude of requirements needed for complete chemical kinetics model validation. Rather, OH* and CH* time histories provide an additional tool to improve the underlying ground-state chemistry. The present study underscores the fact that the high sensitivity of only a few key reactions (within a larger chemical kinetics model) to measured species time histories can be used to make targeted improvements to the mechanism.

Acknowledgments

At CNRS, the research leading to the results herein received funding from the European Research Council under the European Community's Seventh Framework Programme (FP7/2007-2013)/ERC Grant Agreement n° 291049 – 2G-CSafe. Experiments and analyses conducted at Texas A&M University were supported in part by the National Science Foundation, Grant Number CBET-0832561, and by the Texas A&M Engineering Experiment Station (TEES).

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2013.11.008>.

References

- [1] B. Rotavera, E.L. Petersen, Model predictions of higher-order normal alkane ignition from dilute shock-tube experiments, *Shock Waves* 23 (2013) 345–359.
- [2] C.K. Westbrook, W.J. Pitz, O. Herbinet, H.J. Curran, E.J. Silke, A comprehensive detailed chemical kinetic reaction mechanism for combustion of *n*-alkane hydrocarbons from *n*-octane to *n*-hexadecane, *Combust. Flame* 156 (2009) 181–199.
- [3] D.F. Davidson, S.C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R.K. Hanson, Ignition delay time measurements of normal alkanes and simple oxygenates, *J. Propul. Power* 26 (2010) 280–287.
- [4] H.P.S. Shen, J. Steinberg, J. Vanderover, M.A. Oehlschlaeger, A shock tube study of the ignition of *n*-heptane, *n*-decane, *n*-dodecane, and *n*-tetradecane at elevated pressures, *Energy Fuels* 23 (2009) 2482–2489.
- [5] D.C. Horning, D.F. Davidson, R.K. Hanson, Study of the high-temperature autoignition of *n*-Alkane/O₂/Ar mixtures, *J. Propul. Power* 18 (2002) 363–371.
- [6] B. Rotavera, P. Diéart, C. Togbé, P. Dagaut, E.L. Petersen, Oxidation kinetics of *n*-nonane: measurements and modeling of ignition delay times and product concentrations, *Proc. Combust. Inst.* 33 (2011) 175–183.
- [7] A. Mzê-Ahmed, K. Hadj-Ali, P. Dagaut, G. Dayma, Experimental and modeling study of the oxidation kinetics of *n*-Undecane and *n*-Dodecane in a jet-stirred reactor, *Energy Fuels* 26 (2012) 4253–4268.
- [8] A. Ristori, P. Dagaut, M. Cathonnet, The oxidation of *n*-hexadecane: experimental and detailed kinetic modeling, *Combust. Flame* 125 (2001) 1128–1137.
- [9] E. Ranzi, P. Gaffuri, T. Faravelli, P. Dagaut, A wide-range modeling study of *n*-heptane oxidation, *Combust. Flame* 103 (1995) 91–106.
- [10] S.P. Zeppieri, S.D. Klotz, F.L. Dryer, Modeling concepts for larger carbon number alkanes: A partially reduced skeletal mechanism for *n*-decane oxidation and pyrolysis, *Proc. Combust. Inst.* 28 (2000) 1587–1595.
- [11] R.P. Lindstedt, L.Q. Maurice, Detailed kinetic modelling of *n*-heptane combustion, *Combust. Sci. Technol.* 107 (1995) 317–353.
- [12] H.J. Curran, P. Gaffuri, W.J. Pitz, C.K. Westbrook, A comprehensive modeling study of *n*-heptane oxidation, *Combust. Flame* 114 (1998) 149–177.

- [13] S.M. Sarathy, C.K. Westbrook, M. Mehl, W.J. Pitz, C. Togbe, P. Dagaut, H. Wang, M.A. Oehlschlaeger, U. Niemann, K. Seshadri, P.S. Veloo, C. Ji, F.N. Egolfopoulos, T. Lu, Comprehensive chemical kinetic modeling of the oxidation of 2-methylalkanes from C7 to C20, *Combust. Flame* 158 (2011) 2338–2357.
- [14] R.K. Hanson, Applications of quantitative laser sensors to kinetics, propulsion and practical energy systems, *Proc. Combust. Inst.* 33 (2011) 1–40.
- [15] C.A. Taatjes, Uncovering the fundamental chemistry of Alkyl + O₂ reactions via measurements of product formation, *J. Phys. Chem. A* 110 (2006) 4299–4312.
- [16] S.S. Vasu, J. Zádor, D.F. Davidson, R.K. Hanson, D.M. Golden, J.A. Miller, High-temperature measurements and a theoretical study of the reaction of OH with 1,3-Butadiene, *J. Phys. Chem. A* 114 (2010) 8312–8318.
- [17] S.S. Vasu, L.K. Huynh, D.F. Davidson, R.K. Hanson, D.M. Golden, Reactions of OH with butene isomers: measurements of the overall rates and a theoretical study, *J. Phys. Chem. A* 115 (2011) 2549–2556.
- [18] S. Garner, K. Brezinsky, Biologically derived diesel fuel and NO formation: An experimental and chemical kinetic study, Part 1, *Combust. Flame* 158 (2011) 2289–2301.
- [19] S.M. Walton, D.M. Karwat, P.D. Teini, A.M. Gorny, M.S. Wooldridge, Speciation studies of methyl butanoate ignition, *Fuel* 90 (2011) 1796–1804.
- [20] D. Guyot, F. Guethé, B. Schuermans, A. Lacarelle, C.O. Paschereit, CH⁺/OH⁺ chemiluminescence response of an atmospheric premixed flame under varying operating conditions, *ASME Conf. Proc.* 2010 (2010) 933–944.
- [21] D.A. Scott, G.B. King, N.M. Laurendeau, Chemiluminescence-based feedback control of equivalence ratio for a continuous combustor, *J. Propul. Power* 18 (2002) 376–382.
- [22] S.K. Marley, W.L. Roberts, Measurements of laminar burning velocity and Markstein number using high-speed chemiluminescence imaging, *Combust. Flame* 141 (2005) 473–477.
- [23] J. Dec, C. Espey, Chemiluminescence imaging of autoignition in a DI diesel engine, international fall fuels and lubricants meeting and exposition, SAE Technical Paper 982685 (1998).
- [24] M.P. Dunphy, J.M. Simmie, High-temperature oxidation of ethanol. Part 1. - Ignition delays in shock waves, *J. Chem. Soc., Faraday Trans.* 87 (1991) 1691–1696.
- [25] D.F. Cooke, M.G. Dodson, A. Williams, A shock-tube study of the ignition of methanol and ethanol with oxygen, *Combust. Flame* 16 (1971) 233–236.
- [26] L.R. Cancino, M. Fikri, A.A.M. Oliveira, C. Schulz, Ignition delay times of ethanol-containing multi-component gasoline surrogates: Shock-tube experiments and detailed modeling, *Fuel* 90 (2011) 1238–1244.
- [27] F. Buda, R. Bounaceur, V. Warth, P.A. Glaude, R. Fournet, F. Battin-Leclerc, Progress toward a unified detailed kinetic model for the autoignition of alkanes from C4 to C10 between 600 and 1200 K, *Combust. Flame* 142 (2005) 170–186.
- [28] J.M. Hall, E.L. Petersen, An optimized kinetics model for OH chemiluminescence at high temperatures and atmospheric pressures, *Int. J. Chem. Kin.* 38 (2006) 714–724.
- [29] A.B.a.B. Ruscic, 2006.
- [30] C. Muller, V. Michel, G. Scacchi, G.M. Côme, THERGAS – A computer-program for the evaluation of thermochemical data of molecules and free-radicals in the gas phase, *J. Chim. Phys. Phys.-Chim. Biol.* 92 (1995) 1154–1178.
- [31] S.W. Benson, Thermochemical kinetics: methods for the estimation of thermochemical data and rate parameters, 2nd ed., Wiley Interscience, 1976.
- [32] B. Rotavera, Oxidation kinetics of pure and blended methyl octanoate/*n*-nonane/methylcyclohexane: measurements and modeling of OH⁺/CH⁺ chemiluminescence, ignition delay times and laminar flame speeds, in: Ph.D. Dissertation, Texas A&M University, 2012.
- [33] J.d. Vries, C. Aul, A. Barrett, D. Lambe, E.L. Petersen, *Proc. Int. Symp. Shock Waves*, 26 (2007) 171–176.
- [34] C.J. Aul, W.K. Metcalfe, S.M. Burke, H.J. Curran, E.L. Petersen, Ignition and kinetic modeling of methane and ethane fuel blends with oxygen: A design of experiments approach, *Combust. Flame* 160 (2013) 1153–1167.
- [35] E.L. Petersen, M.J.A. Rickard, M.W. Crofton, E.D. Abbey, M.J. Traum, D.M. Kalitan, A facility for gas- and condensed-phase measurements behind shock waves, *Meas. Sci. Technol.* 16 (2005) 1716–1729.
- [36] E. Petersen, M. Kopp, N. Donato, F. Güthe, Assessment of current chemiluminescence kinetics models at engine conditions, *J. Eng. Gas Turbines Power* 134 (2012) 051501.
- [37] C.H. Osorio, A.J. Vissotski, E.L. Petersen, M.S. Mannan, Effect of CF₃Br on C1–C3 ignition and laminar flame speed: Numerical and experimental evaluation, *Combust. Flame* 160 (2013) 1044–1059.
- [38] M.L. Davis, E.L. Petersen, Sensitivity of shock-tube chemiluminescence measurements to the optical setup, in: 29th International symposium on shock waves, Springer, Madison, Wisconsin, 2013.
- [39] E. Friedman, J.H. Mulligan Jr., Ramp Input Response of RC Tree Networks, *Analog Integr. Circ. Sig. Process* 14 (1997) 53–58.
- [40] M.A. Weissman, S.W. Benson, Rate parameters for the reactions of vinyl and butadienyl radicals with hydrogen and acetylene, *J. Phys. Chem.* 92 (1988) 4080–4084.
- [41] A.M. Dean, Predictions of pressure and temperature effects upon radical addition and recombination reactions, *J. Phys. Chem.* 89 (1985) 4600–4608.
- [42] M. Frenklach, H. Wang, M. Goldenberg, G.P. Smith, D.M. Golden, C.T. Bowman, R.K. Hanson, W.C. Gardiner, V. Lissiansky, GRI-Mech 1.2, 1995.
- [43] E.A. Irdam, J.H. Kiefer, L.B. Harding, A.F. Wagner, The formaldehyde decomposition chain mechanism, *Int. J. Chem. Kin.* 25 (1993) 285–303.
- [44] C.D. Wijaya, R. Sumathi, W.H. Green, Thermodynamic properties and kinetic parameters for cyclic ether formation from hydroperoxyalkyl radicals, *J. Phys. Chem. A* 107 (2003) 4908–4920.
- [45] J.A. Manion, R. Louw, Gas-phase hydrogenolysis of chloroethene: rates, products, and computer modelling, *J. Chem. Soc., Perkin Trans.* 2 (1988) 1547–1555.
- [46] W. Tsang, Chemical Kinetic Data Base for Combustion Chemistry Part V, Propene, *J. Phys. Chem. Ref. Data* 20 (1991) 221–273.
- [47] S.S. Vasu, D.F. Davidson, Z. Hong, V. Vasudevan, R.K. Hanson, *Proc. Combust. Inst.* 32 (2009) 173–180.
- [48] E. Ranzi, A. Frassoldati, S. Granata, T. Faravelli, Wide-range kinetic modeling study of the pyrolysis, partial oxidation, and combustion of heavy *n*-alkanes, *Ind. Eng. Chem. Res.* 44 (2005) 5170–5183.
- [49] C.J. Montgomery, Optimized Reduced Chemical Kinetic Mechanisms for Ethylene and JP-8 Combustion, in: 45th AIAA Aerospace Sciences Meeting and Exhibit, American Institute of Aeronautics and Astronautics, 2007.
- [50] H.R. Zhang, E.G. Eddings, A.F. Sarofim, Criteria for selection of components for surrogates of natural gas and transportation fuels, *Proc. Combust. Inst.* 31 (2007) 401–409.
- [51] X. You, F.N. Egolfopoulos, H. Wang, *Proc. Combust. Inst.* 32 (2009) 403–410.
- [52] H. Wang, E. Dames, B. Sirjean, D.A. Sheen, R. Tangko, A. Violi, J.Y.W. Lai, F.N. Egolfopoulos, D.F. Davidson, R.K. Hanson, C.T. Bowman, C.K. Law, W. Tsang, N.P. Cernansky, D.L. Miller, R.P. Lindstedt, JetSurF Version 2.0, 2010.